Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Ed Nunes Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
17399 E Avena RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X203-X190-XX06-XXXX

B. OPERATORS

Nunes, Ed

Operator name: Nunes, Ed

Telephone no.: Landline: ; Cellular: (209) 595-2017

Mailing Address Number and Street	City	State	Zip Code
17399 E Avena RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

Nunes, Everett

Operator name: Nunes, Everett

Telephone no.: Landline: ; Cellular: (209) 595-2017

Mailing Address Number and Street	City	State	Zip Code
17399 Avena RD	Escalon	CA	95320

C. OWNERS

Nunes, Ed

Legal owner name: Nunes, Ed

Telephone no.: Landline: | Cellular: (209) 595-2017

Mailing Address Number and Street	City	State	Zip Code
17399 E Avena RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

Nunes, Everett

Legal owner name: Nunes, Everett

Telephone no.: Landline: | Cellular: (209) 595-2017

Mailing Address Number and Street	City	State	Zip Code
17399 Avena RD	Escalon	CA	95320

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	376	394	0
Number under roof	1,859	294	547	0	0	398
Maximum number	1,879	298	561	387	404	407
Average number	1,859	294	547	376	394	398
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 65,165.27 *tons per reporting period*

Total nitrogen from manure: 819,453.31 *lbs per reporting period*

After ammonia losses (30% loss applied): 573,617.32 *lbs per reporting period*

Total phosphorus from manure: 132,940.26 *lbs per reporting period*

Total potassium from manure: 362,122.47 *lbs per reporting period*

Total salt from manure: 942,915.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 50,756,183 *gallons*

Total nitrogen generated: 213,467.18 *lbs*

Total phosphorus generated: 36,535.00 *lbs*

Total potassium generated: 410,887.05 *lbs*

Total salt generated: 2,542,274.51 *lbs*

	50,756,183	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	50,756,183	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
1	Ground water
2	Ground water
3	Ground water
4	Ground water
5	Ground water

Source Description	Type
8	Ground water
8a	Ground water
9	Ground water
A	Ground water
B	Ground water
C	Ground water
D	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	70	70	2	process wastewater	X203-X190-XX06-XXXX
10	70	70	2	process wastewater	X205-X120-XX06-XXXX X205-X120-XX07-XXXX X205-X120-XX11-XXXX X205-X120-XX12-XXXX
11	40	40	2	process wastewater	X205-X120-XX08-XXXX X205-X120-XX09-XXXX X205-X120-XX13-XXXX X205-X120-XX14-XXXX
12	40	40	2	process wastewater	X205-X120-XX09-XXXX X205-X120-XX10-XXXX X205-X120-XX14-XXXX
13	40	40	2	process wastewater	X203-X190-XX03-XXXX
14	40	40	2	process wastewater	X203-X190-XX03-XXXX
2	30	30	2	process wastewater	X203-X190-XX06-XXXX
3	70	70	2	process wastewater	X203-X190-XX05-XXXX
4	80	80	2	process wastewater	X203-X190-XX05-XXXX
5	45	45	2	process wastewater	X205-X400-XX01-XXXX
6	60	60	2	process wastewater	X205-X400-XX01-XXXX
7	40	40	2	process wastewater	X203-X190-XX07-XXXX
8	35	35	2	process wastewater	X203-X190-XX07-XXXX X203-X190-XX08-XXXX
Totals for areas that were used for application	660	660	26		
Totals for areas that were not used for application					
Land application area totals	660	660	26		

B. CROPS AND HARVESTS

1

Field name: 1

1

10/29/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,096.00 *ton*	Dry-weight		81.4	23,200.00	5,500.00	41,700.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	15.66	135.13	32.03	242.88	693.11

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	2,072.00 *ton*	Dry-weight		62.9	12,000.00	2,800.00	14,400.00		6.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.60	263.56	61.50	316.27	1,458.36

10

Field name: 10

11/13/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,096.00 *ton*	Dry-weight		77.4	13,300.00	3,900.00	20,000.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	15.66	94.12	27.60	141.54	905.86

10

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	2,216.00 *ton*	Dry-weight		65.7	11,200.00	2,900.00	12,300.00		6.41

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	31.66	243.23	62.98	267.12	1,392.05

11

Field name: 11

11/04/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	404.00 *ton*	Dry-weight		76.7	21,400.00	5,600.00	33,800.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	10.10	100.72	26.36	159.08	593.03

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	1,117.00 *ton*	Dry-weight		65.0	11,300.00	2,800.00	15,600.00		6.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	27.93	220.89	54.73	304.94	1,340.96

12

Field name: 12

11/06/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	564.00 *ton*	Dry-weight		67.5	11,800.00	3,000.00	15,100.00		8.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.10	108.15	27.50	138.39	823.93

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,304.00 *ton*	Dry-weight		64.4	9,600.00	2,600.00	8,700.00		4.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	32.60	222.83	60.35	201.94	1,077.00

13

Field name: 13

11/07/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	508.00 *ton*	Dry-weight		80.4	22,300.00	4,800.00	36,800.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	12.70	111.02	23.90	183.21	612.34

13

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,296.00 *ton*	Dry-weight		66.4	12,200.00	2,600.00	12,700.00		5.81

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	32.40	265.63	56.61	276.51	1,265.00

14

Field name: 14

11/05/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	639.00 *ton*	Dry-weight		62.9	17,900.00	3,000.00	15,800.00		6.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	15.98	212.18	35.56	187.28	785.88

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,178.00 *ton*	Dry-weight		66.7	11,700.00	3,000.00	13,200.00		6.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	29.45	229.48	58.84	258.90	1,298.43

2

Field name: 2

10/29/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 30 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	439.00 *ton*	Dry-weight		76.5	14,300.00	4,700.00	30,500.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.63	98.35	32.33	209.77	735.91

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	829.00 *ton*	Dry-weight		64.2	11,100.00	2,000.00	13,900.00		5.27

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	27.63	219.62	39.57	275.02	1,042.69

3

Field name: 3

10/30/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	857.00 *ton*	Dry-weight		82.2	20,700.00	5,300.00	42,300.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	12.24	90.22	23.10	184.36	549.17

3

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	2,219.00 *ton*	Dry-weight		70.1	11,200.00	2,500.00	11,400.00		6.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	31.70	212.31	47.39	216.11	1,144.98

4

Field name: 4

10/31/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 80 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,176.00 *ton*	Dry-weight		62.9	8,300.00	3,200.00	14,600.00		8.71

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.70	90.53	34.90	159.25	950.03

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	2,562.00 *ton*	Dry-weight		70.2	11,300.00	2,700.00	10,300.00		5.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	32.03	215.68	51.53	196.60	1,059.32

5

Field name: 5

10/31/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 45 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	642.00 *ton*	Dry-weight		75.9	16,200.00	3,900.00	23,700.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.27	111.40	26.82	162.97	694.53

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,248.00 *ton*	Dry-weight		67.2	12,100.00	3,000.00	13,300.00		6.01

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	27.73	220.14	54.58	241.97	1,093.40

6

Field name: 6

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 60 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	797.00 *ton*	Dry-weight		76.5	17,400.00	3,700.00	30,300.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	13.28	108.63	23.10	189.17	649.29

6

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,861.00 *ton*	Dry-weight		66.8	12,000.00	2,800.00	11,900.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	31.02	247.14	57.67	245.08	1,173.92

7

Field name: 7

11/06/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 40 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	597.00 *ton*	Dry-weight		76.8	17,200.00	4,500.00	31,600.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	14.93	119.11	31.16	218.84	858.72

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,258.00 *ton*	Dry-weight		70.0	12,100.00	2,400.00	13,300.00		6.23

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	31.45	228.33	45.29	250.97	1,175.60

8

Field name: 8

11/04/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 35 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	403.00 *ton*	Dry-weight		76.6	13,100.00	3,600.00	17,700.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	1,494.00
Total actual harvest content	11.51	70.59	19.40	95.38	630.48

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,059.00 *ton*	Dry-weight		70.3	10,000.00	2,400.00	11,300.00		6.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	1,500.00
Total actual harvest content	30.26	179.73	43.13	203.09	1,108.92

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/29/2022: Wheat, silage, soft dough

Field name: 1

Crop: Wheat, silage, soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	82.68	9.66	152.33	1,326.78	1,030,497.00 *gal*
3	Ground water	9.13	0.00	0.00	347.66	4,942,800.00 *gal*
Application event totals		91.81	9.66	152.33	1,674.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.24	9.52	84.56	563.76	760,284.00 *gal*
3	Ground water	10.63	0.00	0.00	404.60	5,752,418.00 *gal*
Application event totals		82.87	9.52	84.56	968.36	

1 - 06/08/2023: Corn, silage

Field name: 1

Crop: Corn, silage — Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	78.89	13.40	131.48	400.71	1,050,400.00 *gal*
3	Ground water	23.24	0.00	0.00	884.53	12,575,700.00 *gal*
Application event totals		102.13	13.40	131.48	1,285.24	

1 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	22.53	0.00	0.00	857.74	12,194,800.00 *gal*
Application event totals		22.53	0.00	0.00	857.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.35	12.12	118.92	362.41	950,000.00 *gal*
3	Ground water	21.42	0.00	0.00	815.50	11,594,331.00 *gal*
Application event totals		92.77	12.12	118.92	1,177.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.38	12.12	118.96	362.55	950,355.00 *gal*
3	Ground water	20.59	0.00	0.00	783.86	11,144,400.00 *gal*
Application event totals		91.97	12.12	118.96	1,146.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.65	0.00	0.00	785.95	11,174,100.00 *gal*
Application event totals		20.65	0.00	0.00	785.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	21.04	0.00	0.00	800.72	11,384,100.00 *gal*
Application event totals		21.04	0.00	0.00	800.72	

10 - 11/13/2022: Wheat, silage, soft dough

Field name: 10

Crop: Wheat, silage, soft dough

Plant date: 11/13/2022

10 - 11/13/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	61.00	7.12	112.39	978.91	760,312.00 *gal*
8a	Ground water	1.84	0.00	0.00	114.01	5,313,000.00 *gal*
Application event totals		62.84	7.12	112.39	1,092.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	54.18	7.14	63.42	422.81	570,200.00 *gal*
8a	Ground water	0.99	0.00	0.00	61.18	2,851,000.00 *gal*
Application event totals		55.16	7.14	63.42	483.99	

10 - 06/14/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	86.41	14.68	144.01	438.90	1,150,500.00 *gal*
8a	Ground water	4.19	0.00	0.00	260.24	12,127,750.00 *gal*
Application event totals		90.60	14.68	144.01	699.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.79	0.00	0.00	235.02	10,952,400.00 *gal*
Application event totals		3.79	0.00	0.00	235.02	

10 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.86	0.00	0.00	239.31	11,152,200.00 *gal*
Application event totals		3.86	0.00	0.00	239.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	65.55	17.15	196.16	1,515.46	2,090,800.00 *gal*
8a	Ground water	3.51	0.00	0.00	217.63	10,142,050.00 *gal*
Application event totals		69.06	17.15	196.16	1,733.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	53.63	14.03	160.49	1,239.91	1,710,639.00 *gal*
8a	Ground water	3.55	0.00	0.00	220.10	10,257,050.00 *gal*
Application event totals		57.18	14.03	160.49	1,460.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.59	15.59	178.33	1,377.68	1,900,710.00 *gal*
8a	Ground water	3.41	0.00	0.00	211.52	9,857,150.00 *gal*
Application event totals		63.00	15.59	178.33	1,589.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.82	0.00	0.00	236.84	11,037,050.00 *gal*
Application event totals		3.82	0.00	0.00	236.84	

11 - 11/04/2022: Wheat, silage, soft dough

Field name: 11

Crop: Wheat, silage, soft dough

Plant date: 11/04/2022

11 - 11/04/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	63.42	7.41	116.85	1,017.68	451,672.00 *gal*
8a	Ground water	1.86	0.00	0.00	115.76	3,082,500.00 *gal*
Application event totals		65.28	7.41	116.85	1,133.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	61.00	7.12	112.39	978.88	434,448.00 *gal*
8a	Ground water	2.15	0.00	0.00	133.76	3,561,900.00 *gal*
Application event totals		63.15	7.12	112.39	1,112.63	

11 - 06/12/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	97.98	16.64	163.31	497.70	745,500.00 *gal*
8a	Ground water	3.66	0.00	0.00	227.41	6,055,700.00 *gal*
Application event totals		101.65	16.64	163.31	725.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.81	0.00	0.00	236.51	6,298,150.00 *gal*
Application event totals		3.81	0.00	0.00	236.51	

11 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	84.39	14.33	140.65	428.64	642,060.00 *gal*
8a	Ground water	3.94	0.00	0.00	244.57	6,512,800.00 *gal*
Application event totals		88.33	14.33	140.65	673.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.90	0.00	0.00	241.98	6,443,850.00 *gal*
Application event totals		3.90	0.00	0.00	241.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	65.55	17.15	196.16	1,515.45	1,194,732.00 *gal*
8a	Ground water	3.60	0.00	0.00	223.75	5,958,300.00 *gal*
Application event totals		69.16	17.15	196.16	1,739.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.73	0.00	0.00	231.69	6,169,700.00 *gal*
Application event totals		3.73	0.00	0.00	231.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.33	0.00	0.00	206.84	5,508,150.00 *gal*
Application event totals		3.33	0.00	0.00	206.84	

12 - 11/06/2022: Wheat, silage, soft dough

Field name: 12

Crop: Wheat, silage, soft dough

Plant date: 11/06/2022

12 - 11/06/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	62.22	7.27	114.63	998.43	443,125.00 *gal*
8a	Ground water	1.71	0.00	0.00	106.04	2,823,912.00 *gal*
Application event totals		63.93	7.27	114.63	1,104.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.55	9.43	83.76	558.38	430,300.00 *gal*
8a	Ground water	2.34	0.00	0.00	145.02	3,861,900.00 *gal*
Application event totals		73.88	9.43	83.76	703.40	

12 - 06/17/2023: Corn, silage

Field name: 12

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	85.66	14.55	142.76	435.07	651,700.00 *gal*
8a	Ground water	3.72	0.00	0.00	230.84	6,147,250.00 *gal*
Application event totals		89.37	14.55	142.76	665.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.86	0.00	0.00	239.52	6,378,300.00 *gal*
Application event totals		3.86	0.00	0.00	239.52	

12 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	57.10	9.70	95.17	290.04	434,448.00 *gal*
8a	Ground water	3.20	0.00	0.00	198.86	5,295,550.00 *gal*
Application event totals		60.31	9.70	95.17	488.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.79	0.00	0.00	235.44	6,269,700.00 *gal*
Application event totals		3.79	0.00	0.00	235.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.59	15.59	178.33	1,377.68	1,086,120.00 *gal*
8a	Ground water	3.46	0.00	0.00	214.95	5,724,000.00 *gal*
Application event totals		63.06	15.59	178.33	1,592.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	60.54	15.84	181.16	1,399.53	1,103,344.00 *gal*
8a	Ground water	3.76	0.00	0.00	233.40	6,215,400.00 *gal*
Application event totals		64.30	15.84	181.16	1,632.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.61	0.00	0.00	224.18	5,969,700.00 *gal*
Application event totals		3.61	0.00	0.00	224.18	

13 - 11/07/2022: Wheat, silage, soft dough

Field name: 13

Crop: Wheat, silage, soft dough

Plant date: 11/07/2022

13 - 11/07/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	96.24	11.24	177.32	1,544.42	685,448.00 *gal*
5	Ground water	19.77	0.00	0.00	453.18	2,715,300.00 *gal*
Application event totals		116.01	11.24	177.32	1,997.60	

13 - 06/17/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	99.93	16.97	166.54	507.57	760,284.00 *gal*
3	Ground water	19.06	0.00	0.00	725.36	5,892,950.00 *gal*
Application event totals		118.98	16.97	166.54	1,232.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.02	0.00	0.00	762.03	6,190,884.00 *gal*
Application event totals		20.02	0.00	0.00	762.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.37	10.08	98.95	301.55	451,700.00 *gal*
8a	Ground water	3.97	0.00	0.00	246.71	6,569,700.00 *gal*
Application event totals		63.34	10.08	98.95	548.26	

13 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.10	0.00	0.00	765.05	6,215,400.00 *gal*
Application event totals		20.10	0.00	0.00	765.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.51	18.71	213.99	1,653.21	1,303,344.00 *gal*
8a	Ground water	3.92	0.00	0.00	243.28	6,478,300.00 *gal*
Application event totals		75.43	18.71	213.99	1,896.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.89	0.00	0.00	795.29	6,461,100.00 *gal*
Application event totals		20.89	0.00	0.00	795.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	19.12	0.00	0.00	727.90	5,913,600.00 *gal*
Application event totals		19.12	0.00	0.00	727.90	

14 - 11/05/2022: Wheat, silage, soft dough

Field name: 14

Crop: Wheat, silage, soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

14 - 11/05/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	96.10	11.22	177.06	1,542.16	684,448.00 *gal*
3	Ground water	9.13	0.00	0.00	347.59	2,823,912.00 *gal*
Application event totals		105.23	11.22	177.06	1,889.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	97.18	12.80	113.76	758.41	584,448.00 *gal*
3	Ground water	10.89	0.00	0.00	414.44	3,366,972.00 *gal*
Application event totals		108.07	12.80	113.76	1,172.84	

14 - 06/16/2023: Corn, silage

Field name: 14

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	58.59	9.95	97.66	297.62	445,800.00 *gal*
1	Ground water	48.86	0.00	0.00	1,032.33	5,498,076.00 *gal*
Application event totals		107.46	9.95	97.66	1,329.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.04	0.00	0.00	762.91	6,198,076.00 *gal*
Application event totals		20.04	0.00	0.00	762.91	

14 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	21.01	0.00	0.00	799.84	6,498,076.00 *gal*
Application event totals		21.01	0.00	0.00	799.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	35.91	9.39	107.45	830.13	654,448.00 *gal*
3	Ground water	21.08	0.00	0.00	802.30	6,518,076.00 *gal*
Application event totals		56.99	9.39	107.45	1,632.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	31.99	8.37	95.73	739.58	583,060.00 *gal*
3	Ground water	20.46	0.00	0.00	778.91	6,328,076.00 *gal*
Application event totals		52.45	8.37	95.73	1,518.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	19.72	0.00	0.00	750.60	6,098,076.00 *gal*
Application event totals		19.72	0.00	0.00	750.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	18.13	0.00	0.00	690.29	5,608,076.00 *gal*
Application event totals		18.13	0.00	0.00	690.29	

2 - 10/29/2022: Wheat, silage, soft dough

Field name: 2

Crop: Wheat, silage, soft dough

Plant date: 10/29/2022

2 - 10/29/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.17	5.63	88.75	772.98	257,300.00 *gal*
3	Ground water	10.95	0.00	0.00	416.84	2,539,900.00 *gal*
Application event totals		59.12	5.63	88.75	1,189.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	55.80	7.35	65.32	435.49	251,700.00 *gal*
3	Ground water	11.22	0.00	0.00	427.18	2,602,900.00 *gal*
Application event totals		67.02	7.35	65.32	862.67	

2 - 06/07/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	85.65	14.55	142.75	435.06	488,754.00 *gal*
2	Ground water	28.58	0.00	0.00	732.57	4,317,327.00 *gal*
Application event totals		114.23	14.55	142.75	1,167.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.02	0.00	0.00	762.03	4,643,163.00 *gal*
Application event totals		20.02	0.00	0.00	762.03	

2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.50	0.00	0.00	780.32	4,754,600.00 *gal*
Application event totals		20.50	0.00	0.00	780.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.94	0.00	0.00	797.20	4,857,500.00 *gal*
Application event totals		20.94	0.00	0.00	797.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	48.71	12.74	145.76	1,126.04	665,800.00 *gal*
3	Ground water	18.54	0.00	0.00	705.62	4,299,450.00 *gal*
Application event totals		67.25	12.74	145.76	1,831.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.36	0.00	0.00	774.91	4,721,650.00 *gal*
Application event totals		20.36	0.00	0.00	774.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	19.48	0.00	0.00	741.49	4,518,000.00 *gal*
Application event totals		19.48	0.00	0.00	741.49	

3 - 10/30/2022: Wheat, silage, soft dough

Field name: 3

Crop: Wheat, silage, soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 10/30/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	91.50	10.69	168.58	1,468.31	1,140,426.00 *gal*
3	Ground water	9.07	0.00	0.00	345.16	4,907,300.00 *gal*
Application event totals		100.57	10.69	168.58	1,813.48	

3 - 06/02/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	85.65	14.55	142.75	435.06	1,140,426.00 *gal*
3	Ground water	19.67	0.00	0.00	748.66	10,643,976.00 *gal*
Application event totals		105.32	14.55	142.75	1,183.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.02	0.00	0.00	762.03	10,834,047.00 *gal*
Application event totals		20.02	0.00	0.00	762.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.72	0.00	0.00	788.77	11,214,189.00 *gal*
Application event totals		20.72	0.00	0.00	788.77	

3 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	57.11	9.70	95.18	290.08	760,400.00 *gal*
3	Ground water	21.07	0.00	0.00	802.13	11,404,260.00 *gal*
Application event totals		78.18	9.70	95.18	1,092.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.37	0.00	0.00	775.40	11,024,118.00 *gal*
Application event totals		20.37	0.00	0.00	775.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.02	0.00	0.00	762.03	10,834,047.00 *gal*
Application event totals		20.02	0.00	0.00	762.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	18.51	0.00	0.00	704.56	10,017,050.00 *gal*
Application event totals		18.51	0.00	0.00	704.56	

4 - 10/31/2022: Wheat, silage, soft dough

Field name: 4

Crop: Wheat, silage, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 10/31/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	96.10	11.22	177.06	1,542.17	1,368,900.00 *gal*
3	Ground water	9.36	0.00	0.00	356.41	5,791,100.00 *gal*
Application event totals		105.46	11.22	177.06	1,898.58	

4 - 06/03/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.70	12.35	121.16	369.25	1,106,200.00 *gal*
3	Ground water	19.67	0.00	0.00	748.66	12,164,544.00 *gal*
Application event totals		92.36	12.35	121.16	1,117.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	19.55	0.00	0.00	744.14	12,091,100.00 *gal*
Application event totals		19.55	0.00	0.00	744.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	72.51	12.32	120.85	368.31	1,103,400.00 *gal*
3	Ground water	21.07	0.00	0.00	802.13	13,033,440.00 *gal*
Application event totals		93.59	12.32	120.85	1,170.45	

4 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	21.40	0.00	0.00	814.44	13,233,440.00 *gal*
Application event totals		21.40	0.00	0.00	814.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	20.37	0.00	0.00	775.40	12,598,992.00 *gal*
Application event totals		20.37	0.00	0.00	775.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	21.99	0.00	0.00	836.94	13,598,992.00 *gal*
Application event totals		21.99	0.00	0.00	836.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3	Ground water	17.26	0.00	0.00	656.92	10,673,900.00 *gal*
Application event totals		17.26	0.00	0.00	656.92	

5 - 10/31/2022: Wheat, silage, soft dough

Field name: 5

Crop: Wheat, silage, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	59.59	6.96	109.80	956.35	477,508.00 *gal*
4	Ground water	9.42	0.00	0.00	342.61	3,299,089.00 *gal*
Application event totals		69.02	6.96	109.80	1,298.96	

5 - 10/31/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	64.85	8.54	75.91	506.09	438,754.00 *gal*
4	Ground water	11.17	0.00	0.00	406.05	3,910,032.00 *gal*
Application event totals		76.01	8.54	75.91	912.14	

5 - 06/04/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	99.93	16.97	166.54	507.57	855,319.00 *gal*
4	Ground water	19.66	0.00	0.00	714.77	6,882,750.00 *gal*
Application event totals		119.58	16.97	166.54	1,222.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	20.24	0.00	0.00	735.97	7,086,933.00 *gal*
Application event totals		20.24	0.00	0.00	735.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	20.59	0.00	0.00	748.66	7,209,121.00 *gal*
Application event totals		20.59	0.00	0.00	748.66	

5 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	20.94	0.00	0.00	761.35	7,331,310.00 *gal*
Application event totals		20.94	0.00	0.00	761.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	47.67	12.47	142.66	1,102.14	977,508.00 *gal*
4	Ground water	21.28	0.00	0.00	773.81	7,451,310.00 *gal*
Application event totals		68.95	12.47	142.66	1,875.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	19.89	0.00	0.00	723.28	6,964,744.00 *gal*
Application event totals		19.89	0.00	0.00	723.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	19.29	0.00	0.00	701.47	6,754,745.00 *gal*
Application event totals		19.29	0.00	0.00	701.47	

6 - 11/02/2022: Wheat, silage, soft dough

Field name: 6

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 11/02/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	63.42	7.41	116.85	1,017.68	677,508.00 *gal*
4	Ground water	9.07	0.00	0.00	329.92	4,235,868.00 *gal*
Application event totals		72.49	7.41	116.85	1,347.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.13	8.98	79.75	531.68	614,590.00 *gal*
8a	Ground water	2.14	0.00	0.00	133.03	5,313,600.00 *gal*
Application event totals		70.27	8.98	79.75	664.71	

6 - 06/14/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	85.65	14.55	142.75	435.06	977,508.00 *gal*
4	Ground water	19.54	0.00	0.00	710.59	9,123,408.00 *gal*
Application event totals		105.19	14.55	142.75	1,145.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4	Ground water	20.24	0.00	0.00	735.97	9,449,244.00 *gal*
Application event totals		20.24	0.00	0.00	735.97	

6 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	71.38	12.12	118.96	362.55	814,590.00 *gal*
8a	Ground water	4.01	0.00	0.00	248.80	9,937,998.00 *gal*
Application event totals		75.38	12.12	118.96	611.34	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.94	0.00	0.00	244.72	9,775,080.00 *gal*
Application event totals		3.94	0.00	0.00	244.72	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	58.86	15.40	176.14	1,360.77	1,609,180.00 *gal*
8a	Ground water	3.74	0.00	0.00	232.12	9,272,000.00 *gal*
Application event totals		62.60	15.40	176.14	1,592.89	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	47.67	12.47	142.66	1,102.14	1,303,344.00 *gal*
8a	Ground water	3.68	0.00	0.00	228.69	9,135,000.00 *gal*
Application event totals		51.36	12.47	142.66	1,330.84	

09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.55	0.00	0.00	220.54	8,809,100.00 *gal*
Application event totals		3.55	0.00	0.00	220.54	

7 - 11/06/2022: Wheat, silage, soft dough

Field name: 7

Crop: Wheat, silage, soft dough

Plant date: 11/06/2022

7 - 11/06/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	61.00	7.12	112.39	978.88	434,448.00 *gal*
3	Ground water	8.90	0.00	0.00	338.89	2,753,256.00 *gal*
Application event totals		69.90	7.12	112.39	1,317.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	70.74	9.32	82.81	552.08	425,448.00 *gal*
3	Ground water	11.52	0.00	0.00	438.43	3,561,900.00 *gal*
Application event totals		82.26	9.32	82.81	990.51	

7 - 06/11/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	114.20	19.40	190.34	580.07	868,896.00 *gal*
3	Ground water	19.32	0.00	0.00	735.29	5,973,660.00 *gal*
Application event totals		133.52	19.40	190.34	1,315.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.68	0.00	0.00	228.40	6,082,272.00 *gal*
Application event totals		3.68	0.00	0.00	228.40	

7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.88	0.00	0.00	240.64	6,408,108.00 *gal*
Application event totals		3.88	0.00	0.00	240.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.94	0.00	0.00	244.72	6,516,720.00 *gal*
Application event totals		3.94	0.00	0.00	244.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	76.05	19.90	227.58	1,758.21	1,386,120.00 *gal*
8a	Ground water	3.75	0.00	0.00	232.48	6,190,884.00 *gal*
Application event totals		79.80	19.90	227.58	1,990.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	68.92	18.03	206.24	1,593.31	1,256,120.00 *gal*
8a	Ground water	3.61	0.00	0.00	224.33	5,973,660.00 *gal*
Application event totals		72.54	18.03	206.24	1,817.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.61	0.00	0.00	224.33	5,973,660.00 *gal*
Application event totals		3.61	0.00	0.00	224.33	

8 - 11/04/2022: Wheat, silage, soft dough

Field name: 8

Crop: Wheat, silage, soft dough

Plant date: 11/04/2022

8 - 11/04/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	84.27	9.84	155.27	1,352.35	525,178.00 *gal*
8a	Ground water	1.71	0.00	0.00	106.04	2,470,923.00 *gal*
Application event totals		85.98	9.84	155.27	1,458.39	

8 - 06/07/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	87.15	14.80	145.25	442.68	580,200.00 *gal*
8a	Ground water	3.68	0.00	0.00	228.40	5,321,988.00 *gal*
Application event totals		90.83	14.80	145.25	671.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
5	Ground water	25.31	0.00	0.00	580.10	3,041,250.00 *gal*
8a	Ground water	2.17	0.00	0.00	134.81	3,141,250.00 *gal*
Application event totals		27.48	0.00	0.00	714.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.78	0.00	0.00	234.80	5,471,000.00 *gal*
Application event totals		3.78	0.00	0.00	234.80	

8 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.94	0.00	0.00	244.78	5,703,650.00 *gal*
Application event totals		3.94	0.00	0.00	244.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	55.83	14.60	167.06	1,290.60	890,284.00 *gal*
8a	Ground water	3.80	0.00	0.00	235.78	5,493,750.00 *gal*
Application event totals		59.63	14.60	167.06	1,526.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon	Process wastewater	41.72	10.91	124.83	964.37	665,248.00 *gal*
8a	Ground water	3.85	0.00	0.00	239.09	5,571,000.00 *gal*
Application event totals		45.57	10.91	124.83	1,203.47	

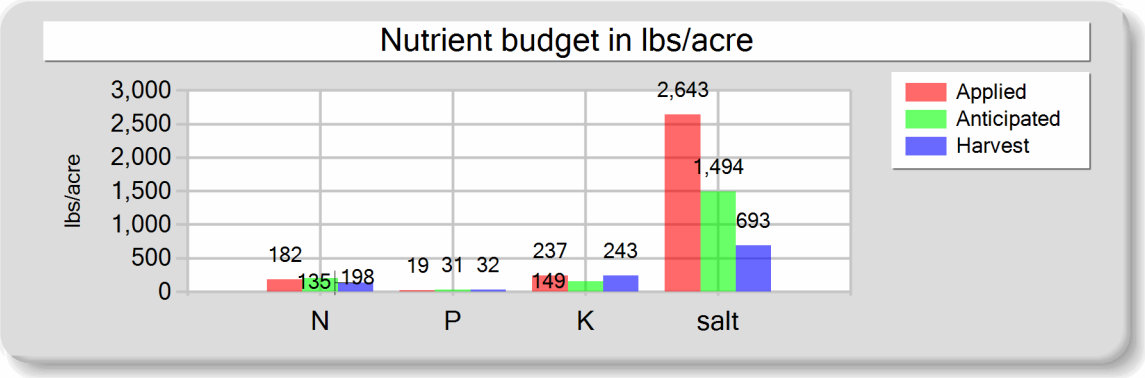
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
8a	Ground water	3.68	0.00	0.00	228.40	5,321,988.00 *gal*
Application event totals		3.68	0.00	0.00	228.40	

B. NUTRIENT BUDGET

1 - 10/29/2022: Wheat, silage, soft dough

Field name: 1 Crop: Wheat, silage, soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.92	19.17	236.90	1,890.54
Fresh water	19.76	0.00	0.00	752.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.68	19.17	236.90	2,642.80
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	135.13	32.03	242.88	693.11
Nutrient balance	46.55	-12.86	-5.98	1,949.69
Applied to removed ratio	1.34	0.60	0.98	3.81

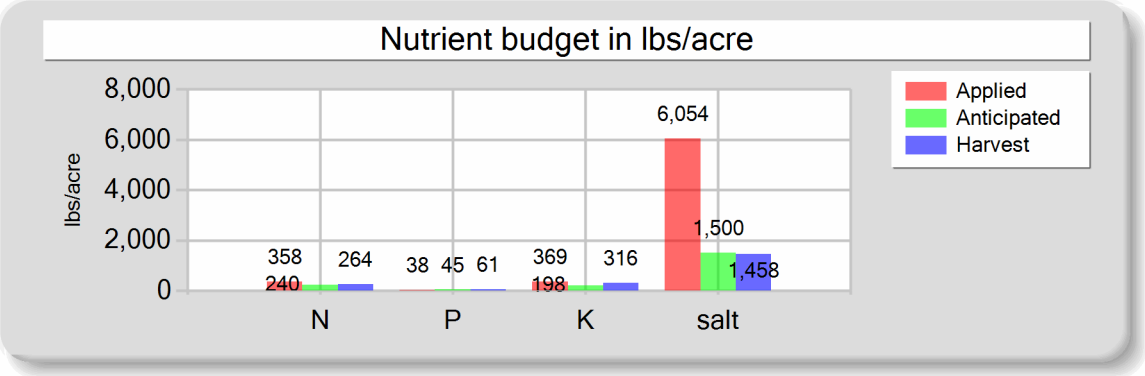
Fresh water applied	
10,695,218.00	*gallons*
393.87	*acre-inches*
5.63	*inches/acre*

Process wastewater applied	
1,790,781.00	*gallons*
65.95	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/08/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.62	37.64	369.36	1,125.67
Fresh water	129.47	0.00	0.00	4,928.29
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	358.09	37.64	369.36	6,053.96
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	263.56	61.50	316.27	1,458.36
Nutrient balance	94.53	-23.86	53.09	4,595.61
Applied to removed ratio	1.36	0.61	1.17	4.15

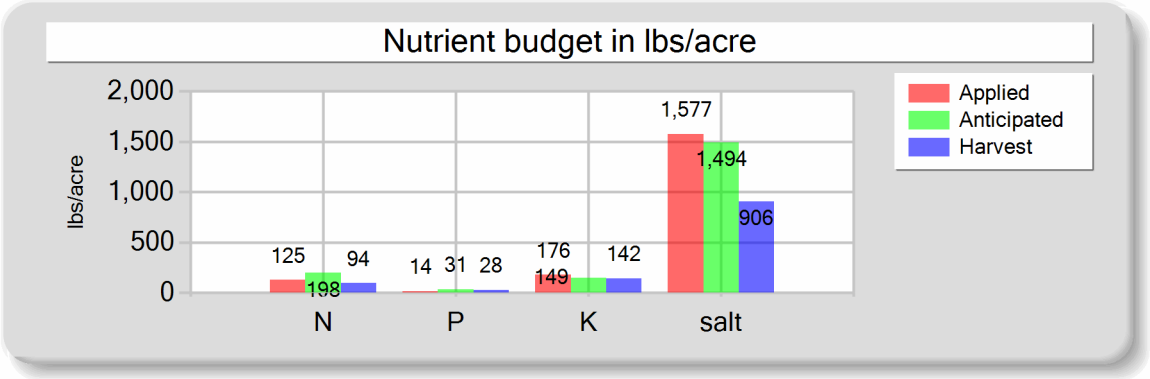
Fresh water applied	
70,067,431.00	*gallons*
2,580.35	*acre-inches*
36.86	*inches/acre*

Process wastewater applied	
2,950,755.00	*gallons*
108.67	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 11/13/2022: Wheat, silage, soft dough

Field name: 10 Crop: Wheat, silage, soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	115.18	14.26	175.82	1,401.72
Fresh water	2.82	0.00	0.00	175.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	125.00	14.26	175.82	1,576.91
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	94.12	27.60	141.54	905.86
Nutrient balance	30.88	-13.34	34.27	671.05
Applied to removed ratio	1.33	0.52	1.24	1.74

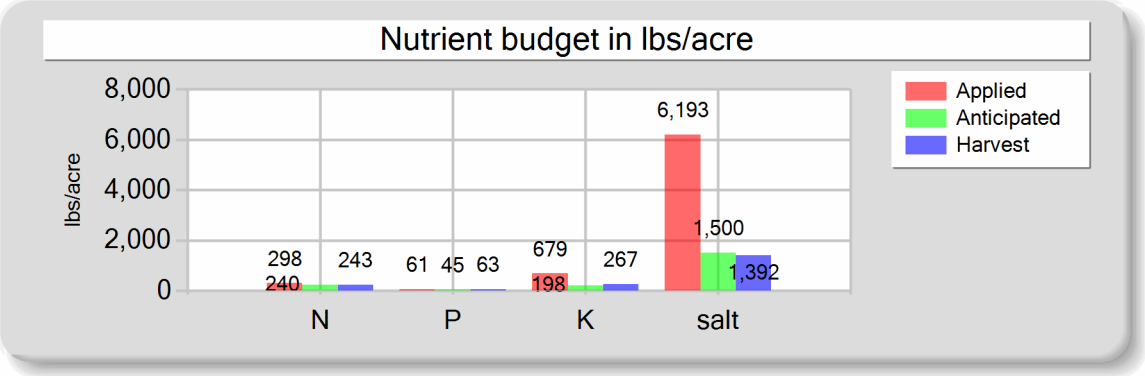
Fresh water applied	
8,164,000.00	*gallons*
300.65	*acre-inches*
4.30	*inches/acre*

Process wastewater applied	
1,330,512.00	*gallons*
49.00	*acre-inches*
0.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/14/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.19	61.44	679.00	4,571.95
Fresh water	26.11	0.00	0.00	1,620.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.30	61.44	679.00	6,192.62
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	243.23	62.98	267.12	1,392.05
Nutrient balance	55.07	-1.53	411.88	4,800.57
Applied to removed ratio	1.23	0.98	2.54	4.45

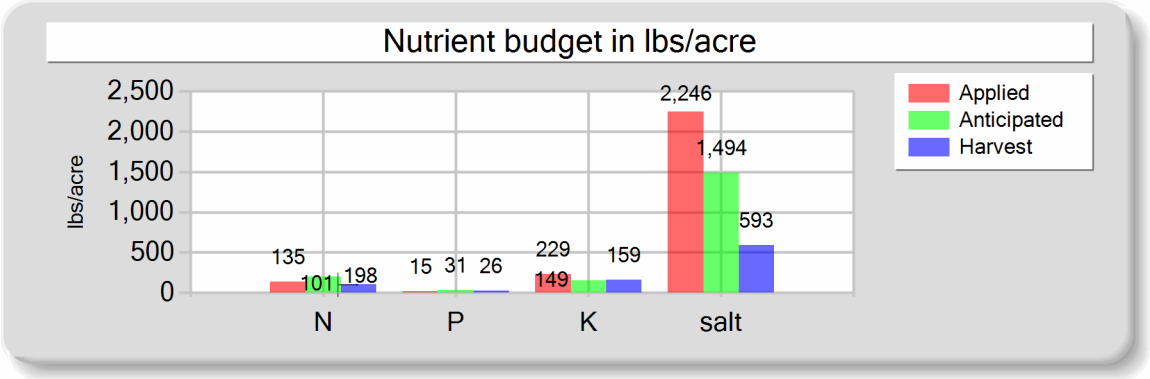
Fresh water applied	
75,525,650.00	*gallons*
2,781.35	*acre-inches*
39.73	*inches/acre*

Process wastewater applied	
6,852,649.00	*gallons*
252.36	*acre-inches*
3.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/04/2022: Wheat, silage, soft dough

Field name: 11 Crop: Wheat, silage, soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	124.42	14.53	229.23	1,996.56
Fresh water	4.02	0.00	0.00	249.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	135.44	14.53	229.23	2,246.08
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	100.72	26.36	159.08	593.03
Nutrient balance	34.71	-11.83	70.15	1,653.04
Applied to removed ratio	1.34	0.55	1.44	3.79

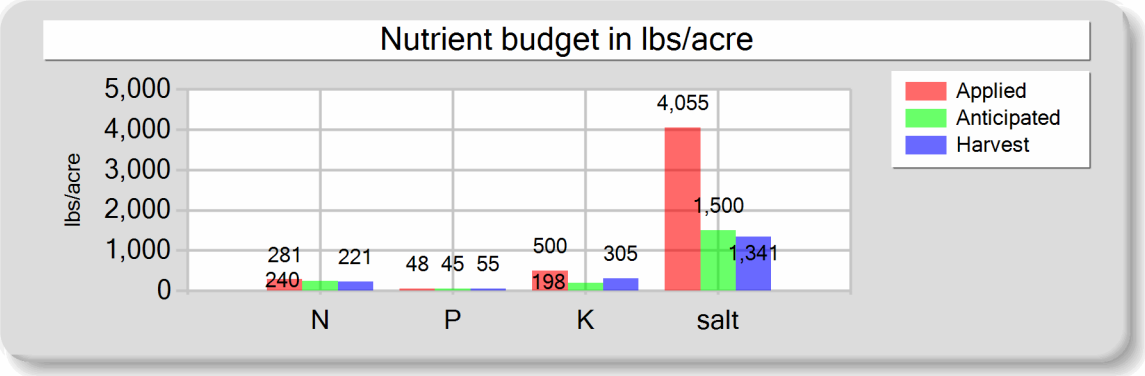
Fresh water applied	
6,644,400.00	*gallons*
244.69	*acre-inches*
6.12	*inches/acre*

Process wastewater applied	
886,120.00	*gallons*
32.63	*acre-inches*
0.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 06/12/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	247.93	48.12	500.11	2,441.78
Fresh water	25.98	0.00	0.00	1,612.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.91	48.12	500.11	4,054.53
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	220.89	54.73	304.94	1,340.96
Nutrient balance	60.02	-6.61	195.17	2,713.58
Applied to removed ratio	1.27	0.88	1.64	3.02

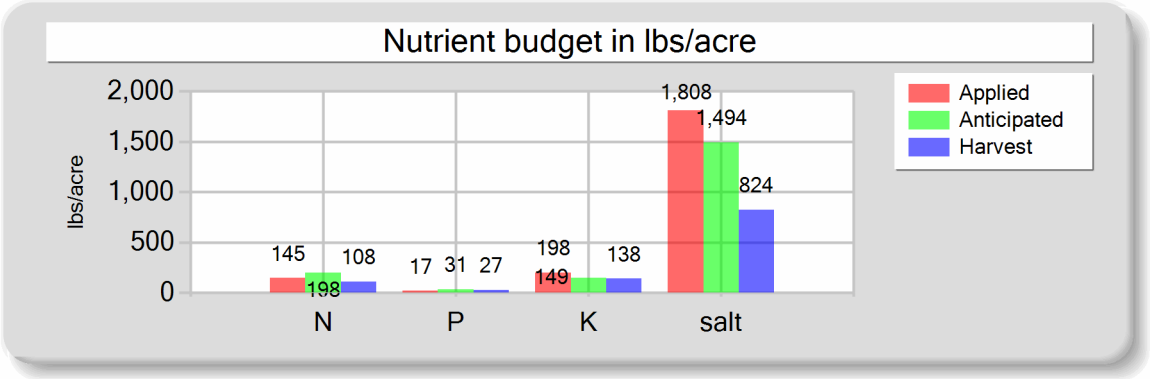
Fresh water applied	
42,946,650.00	*gallons*
1,581.58	*acre-inches*
39.54	*inches/acre*

Process wastewater applied	
2,582,292.00	*gallons*
95.10	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 11/06/2022: Wheat, silage, soft dough

Field name: 12 Crop: Wheat, silage, soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	133.76	16.69	198.39	1,556.80
Fresh water	4.04	0.00	0.00	251.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	144.81	16.69	198.39	1,807.87
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	108.15	27.50	138.39	823.93
Nutrient balance	36.66	-10.80	60.00	983.94
Applied to removed ratio	1.34	0.61	1.43	2.19

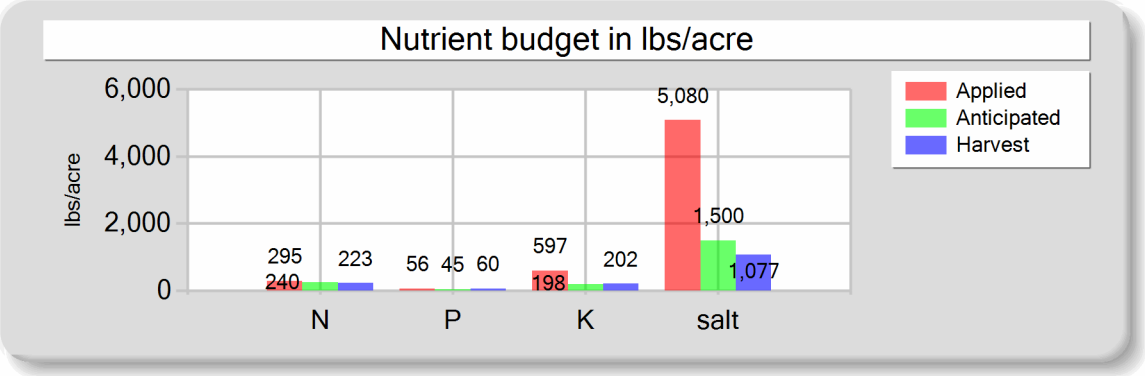
Fresh water applied	
6,685,812.00	*gallons*
246.22	*acre-inches*
6.16	*inches/acre*

Process wastewater applied	
873,425.00	*gallons*
32.17	*acre-inches*
0.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 06/17/2023: Corn, silage

Field name: 12 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.89	55.67	597.41	3,502.32
Fresh water	25.41	0.00	0.00	1,577.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.30	55.67	597.41	5,079.52
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	222.83	60.35	201.94	1,077.00
Nutrient balance	72.47	-4.68	395.47	4,002.52
Applied to removed ratio	1.33	0.92	2.96	4.72

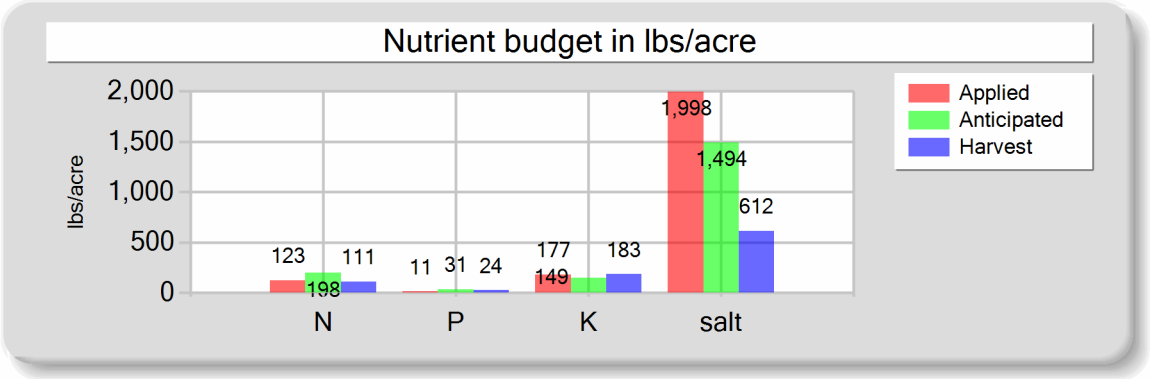
Fresh water applied	
41,999,900.00	*gallons*
1,546.71	*acre-inches*
38.67	*inches/acre*

Process wastewater applied	
3,275,612.00	*gallons*
120.63	*acre-inches*
3.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 11/07/2022: Wheat, silage, soft dough

Field name: 13 Crop: Wheat, silage, soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	96.24	11.24	177.32	1,544.42
Fresh water	19.77	0.00	0.00	453.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	123.01	11.24	177.32	1,997.60
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	111.02	23.90	183.21	612.34
Nutrient balance	11.99	-12.66	-5.88	1,385.26
Applied to removed ratio	1.11	0.47	0.97	3.26

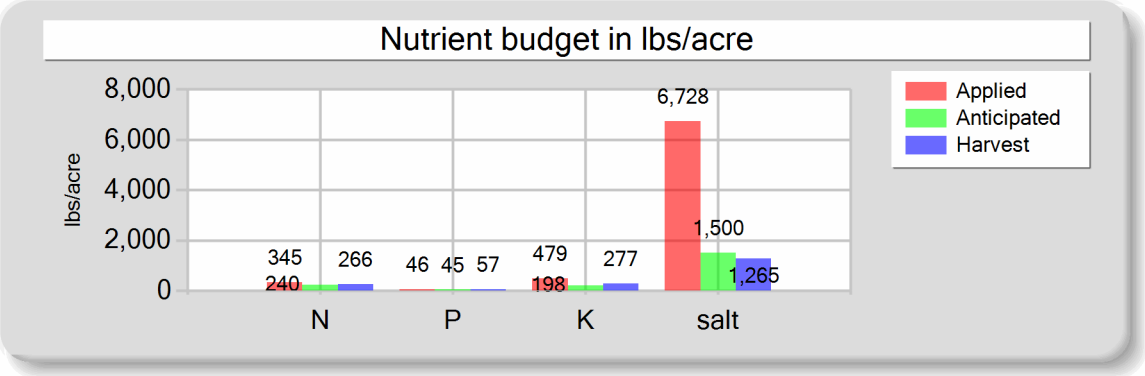
Fresh water applied	
2,715,300.00	*gallons*
100.00	*acre-inches*
2.50	*inches/acre*

Process wastewater applied	
685,448.00	*gallons*
25.24	*acre-inches*
0.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 06/17/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.81	45.76	479.49	2,462.33
Fresh water	107.08	0.00	0.00	4,265.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.89	45.76	479.49	6,727.94
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	265.63	56.61	276.51	1,265.00
Nutrient balance	79.26	-10.85	202.97	5,462.94
Applied to removed ratio	1.30	0.81	1.73	5.32

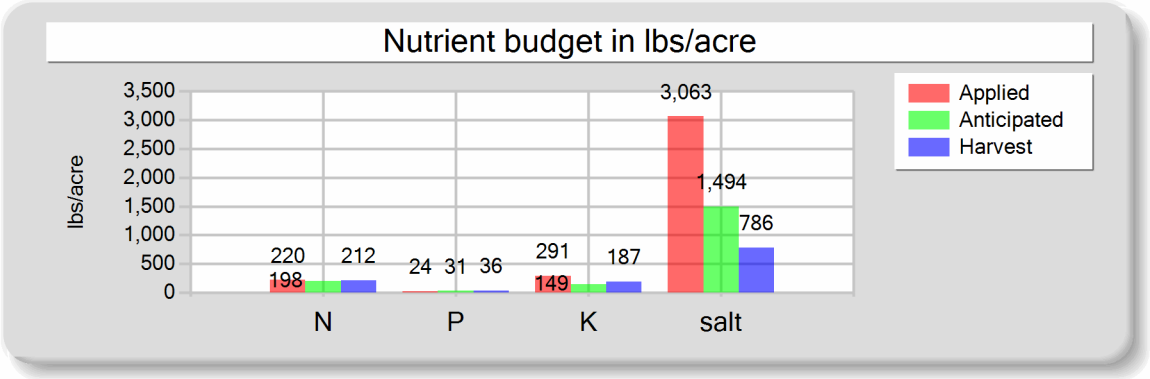
Fresh water applied	
43,721,934.00	*gallons*
1,610.13	*acre-inches*
40.25	*inches/acre*

Process wastewater applied	
2,515,328.00	*gallons*
92.63	*acre-inches*
2.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 11/05/2022: Wheat, silage, soft dough

Field name: 14 Crop: Wheat, silage, soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.28	24.03	290.82	2,300.57
Fresh water	20.02	0.00	0.00	762.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.30	24.03	290.82	3,062.60
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	212.18	35.56	187.28	785.88
Nutrient balance	8.12	-11.53	103.54	2,276.72
Applied to removed ratio	1.04	0.68	1.55	3.90

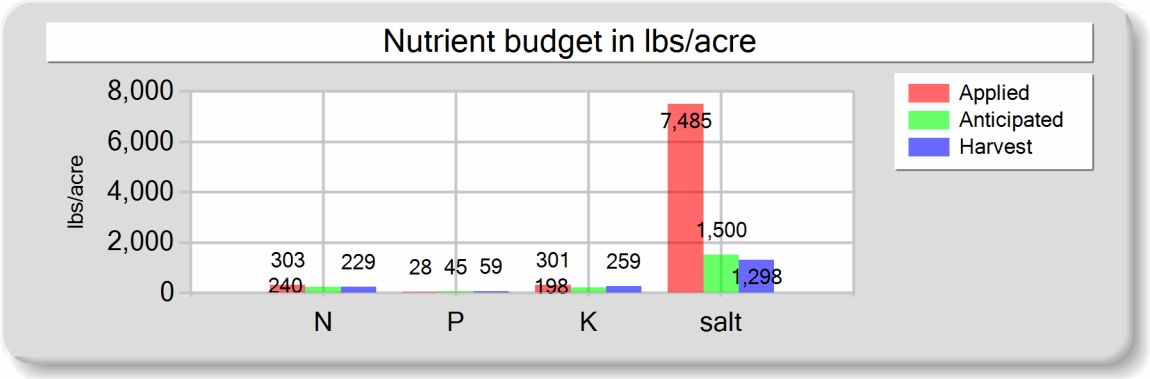
Fresh water applied	
6,190,884.00	*gallons*
227.99	*acre-inches*
5.70	*inches/acre*

Process wastewater applied	
1,268,896.00	*gallons*
46.73	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 06/16/2023: Corn, silage

Field name: 14 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	126.49	27.71	300.84	1,867.32
Fresh water	169.31	0.00	0.00	5,617.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.81	27.71	300.84	7,484.52
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	229.48	58.84	258.90	1,298.43
Nutrient balance	73.33	-31.13	41.94	6,186.09
Applied to removed ratio	1.32	0.47	1.16	5.76

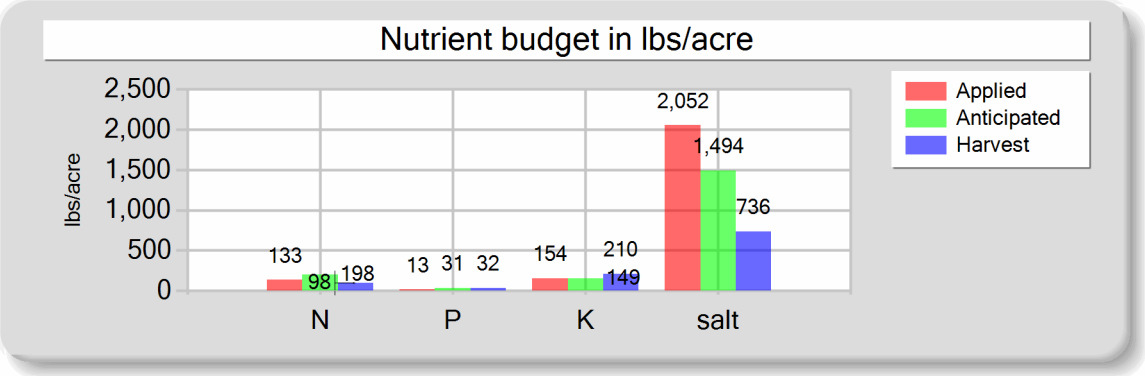
Fresh water applied	
42,746,532.00	*gallons*
1,574.21	*acre-inches*
39.36	*inches/acre*

Process wastewater applied	
1,683,308.00	*gallons*
61.99	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/29/2022: Wheat, silage, soft dough

Field name: 2 Crop: Wheat, silage, soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	103.97	12.98	154.07	1,208.47
Fresh water	22.17	0.00	0.00	844.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	133.14	12.98	154.07	2,052.50
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	98.35	32.33	209.77	735.91
Nutrient balance	34.79	-19.35	-55.70	1,316.59
Applied to removed ratio	1.35	0.40	0.73	2.79

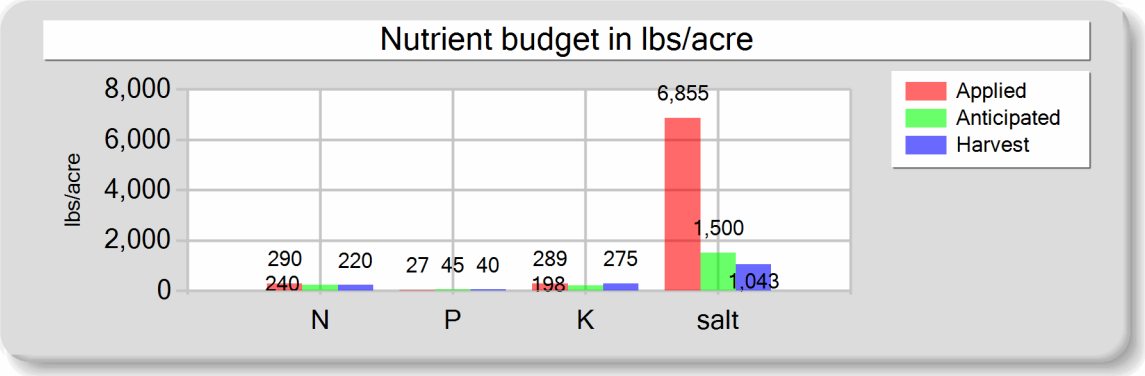
Fresh water applied	
5,142,800.00	*gallons*
189.39	*acre-inches*
6.31	*inches/acre*

Process wastewater applied	
509,000.00	*gallons*
18.74	*acre-inches*
0.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/07/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	134.36	27.29	288.51	1,561.09
Fresh water	148.42	0.00	0.00	5,294.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.78	27.29	288.51	6,855.23
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	219.62	39.57	275.02	1,042.69
Nutrient balance	70.16	-12.28	13.49	5,812.53
Applied to removed ratio	1.32	0.69	1.05	6.57

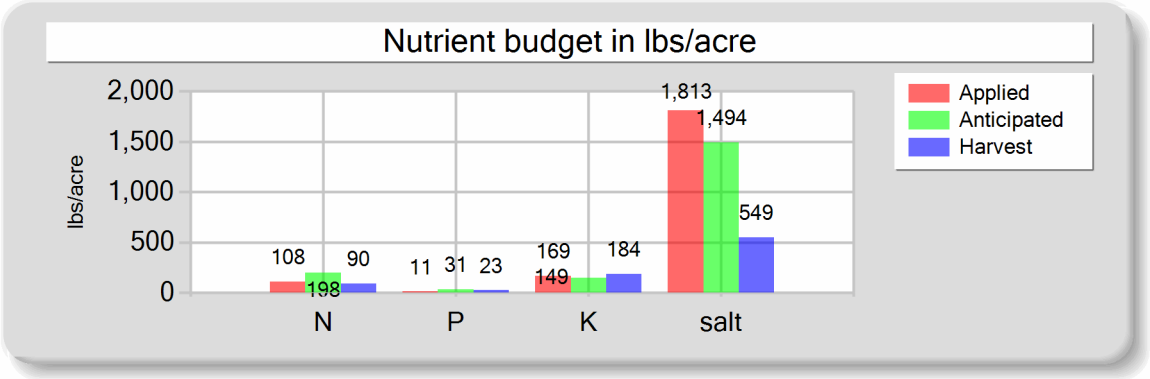
Fresh water applied	
32,111,690.00	*gallons*
1,182.56	*acre-inches*
39.42	*inches/acre*

Process wastewater applied	
1,154,554.00	*gallons*
42.52	*acre-inches*
1.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/30/2022: Wheat, silage, soft dough

Field name: 3 Crop: Wheat, silage, soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	91.50	10.69	168.58	1,468.31
Fresh water	9.07	0.00	0.00	345.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.57	10.69	168.58	1,813.48
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	90.22	23.10	184.36	549.17
Nutrient balance	17.35	-12.41	-15.78	1,264.31
Applied to removed ratio	1.19	0.46	0.91	3.30

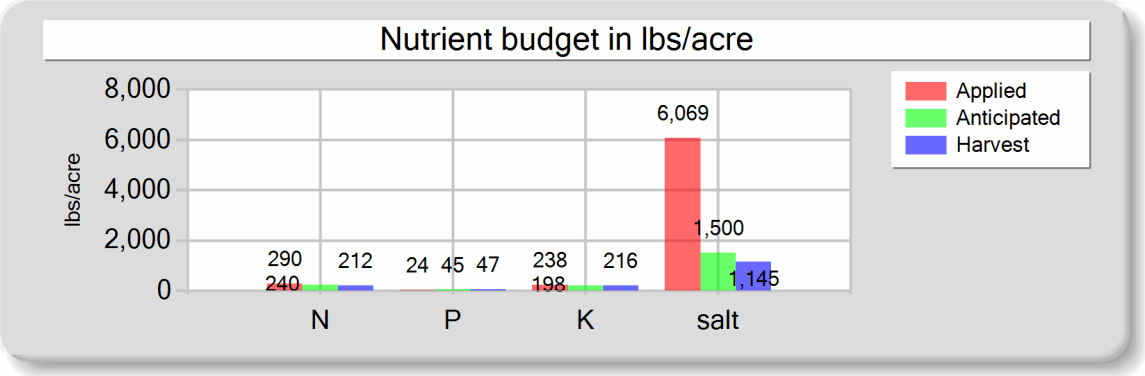
Fresh water applied	
4,907,300.00	*gallons*
180.72	*acre-inches*
2.58	*inches/acre*

Process wastewater applied	
1,140,426.00	*gallons*
42.00	*acre-inches*
0.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/02/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	142.76	24.25	237.94	725.14
Fresh water	140.38	0.00	0.00	5,343.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.14	24.25	237.94	6,068.72
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	212.31	47.39	216.11	1,144.98
Nutrient balance	77.83	-23.14	21.83	4,923.74
Applied to removed ratio	1.37	0.51	1.10	5.30

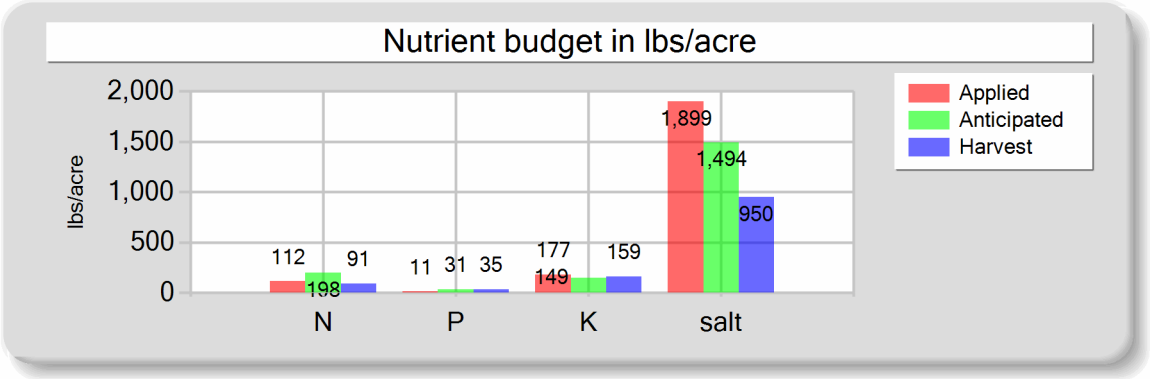
Fresh water applied	
75,971,687.00	*gallons*
2,797.78	*acre-inches*
39.97	*inches/acre*

Process wastewater applied	
1,900,826.00	*gallons*
70.00	*acre-inches*
1.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/31/2022: Wheat, silage, soft dough

Field name: 4 Crop: Wheat, silage, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	96.10	11.22	177.06	1,542.17
Fresh water	9.36	0.00	0.00	356.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	112.46	11.22	177.06	1,898.58
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	90.53	34.90	159.25	950.03
Nutrient balance	21.93	-23.68	17.82	948.54
Applied to removed ratio	1.24	0.32	1.11	2.00

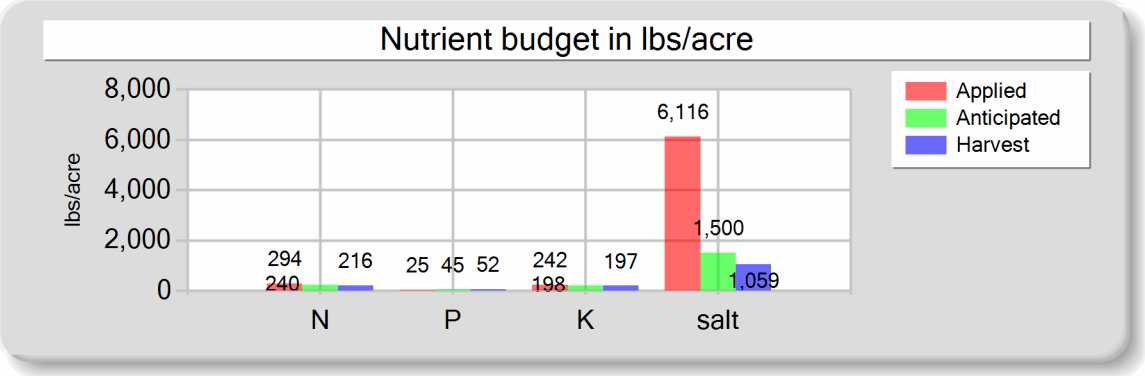
Fresh water applied	
5,791,100.00	*gallons*
213.27	*acre-inches*
2.67	*inches/acre*

Process wastewater applied	
1,368,900.00	*gallons*
50.41	*acre-inches*
0.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/03/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	145.21	24.66	242.01	737.56
Fresh water	141.30	0.00	0.00	5,378.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.51	24.66	242.01	6,116.20
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	215.68	51.53	196.60	1,059.32
Nutrient balance	77.83	-26.87	45.42	5,056.88
Applied to removed ratio	1.36	0.48	1.23	5.77

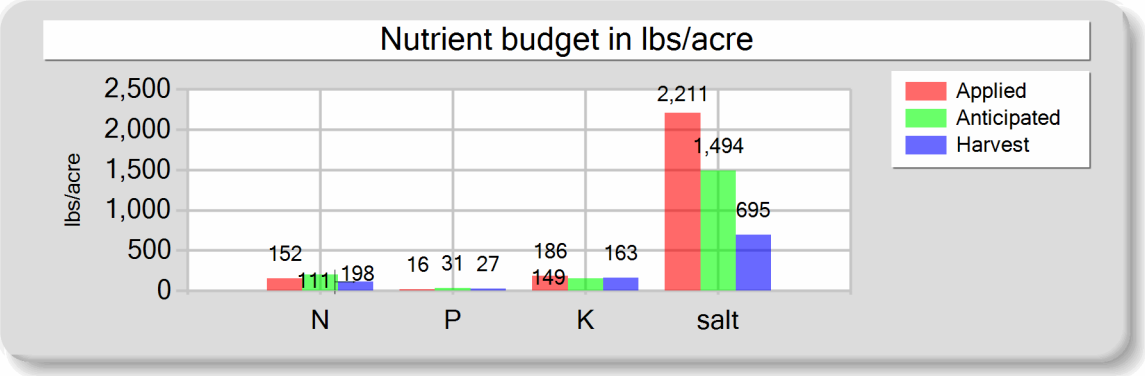
Fresh water applied	
87,394,408.00	*gallons*
3,218.44	*acre-inches*
40.23	*inches/acre*

Process wastewater applied	
2,209,600.00	*gallons*
81.37	*acre-inches*
1.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/31/2022: Wheat, silage, soft dough

Field name: 5 Crop: Wheat, silage, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	124.44	15.50	185.72	1,462.44
Fresh water	20.59	0.00	0.00	748.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.03	15.50	185.72	2,211.10
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	111.40	26.82	162.97	694.53
Nutrient balance	40.63	-11.32	22.74	1,516.57
Applied to removed ratio	1.36	0.58	1.14	3.18

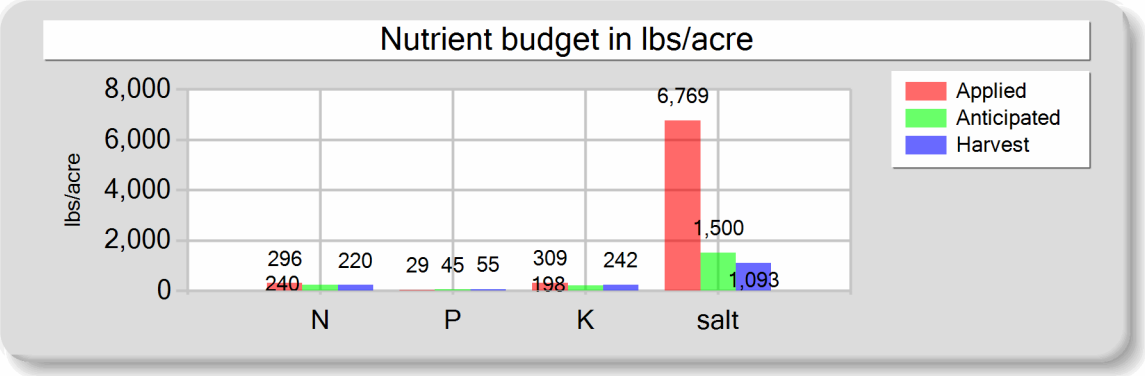
Fresh water applied	
7,209,121.00	*gallons*
265.49	*acre-inches*
5.90	*inches/acre*

Process wastewater applied	
916,262.00	*gallons*
33.74	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/04/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	147.60	29.44	309.21	1,609.71
Fresh water	141.88	0.00	0.00	5,159.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.48	29.44	309.21	6,769.02
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	220.14	54.58	241.97	1,093.40
Nutrient balance	76.35	-25.14	67.24	5,675.61
Applied to removed ratio	1.35	0.54	1.28	6.19

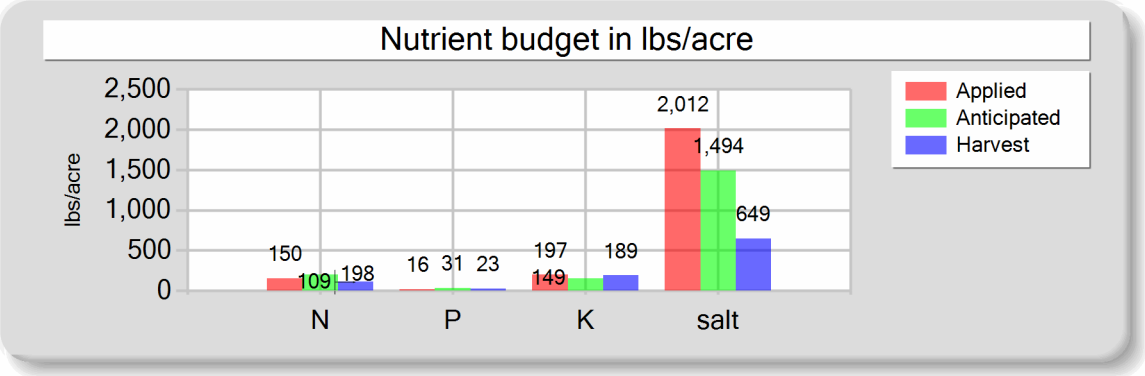
Fresh water applied	
49,680,913.00	*gallons*
1,829.58	*acre-inches*
40.66	*inches/acre*

Process wastewater applied	
1,832,827.00	*gallons*
67.50	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/02/2022: Wheat, silage, soft dough

Field name: 6 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	131.54	16.38	196.60	1,549.37
Fresh water	11.22	0.00	0.00	462.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.76	16.38	196.60	2,012.31
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	108.63	23.10	189.17	649.29
Nutrient balance	41.13	-6.72	7.43	1,363.02
Applied to removed ratio	1.38	0.71	1.04	3.10

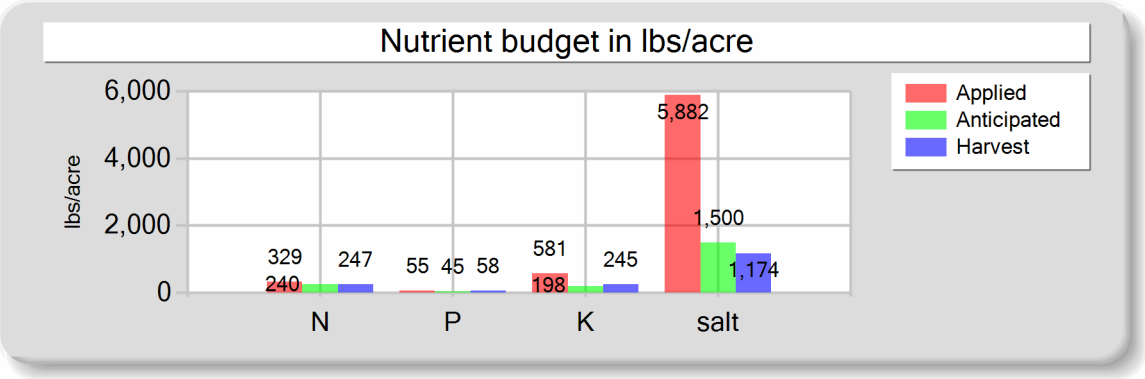
Fresh water applied	
9,549,468.00	*gallons*
351.67	*acre-inches*
5.86	*inches/acre*

Process wastewater applied	
1,292,098.00	*gallons*
47.58	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/14/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	263.57	54.54	580.51	3,260.51
Fresh water	58.71	0.00	0.00	2,621.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.27	54.54	580.51	5,881.94
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	247.14	57.67	245.08	1,173.92
Nutrient balance	82.13	-3.13	335.43	4,708.03
Applied to removed ratio	1.33	0.95	2.37	5.01

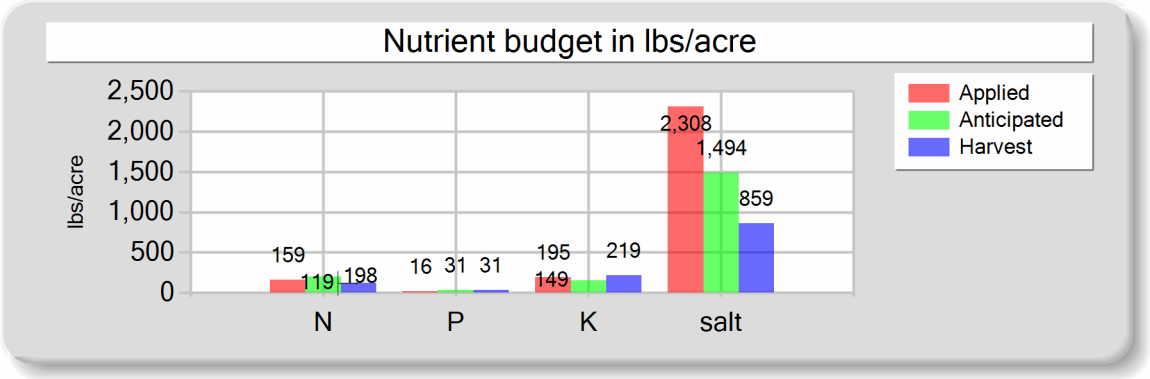
Fresh water applied	
65,501,830.00	*gallons*
2,412.21	*acre-inches*
40.20	*inches/acre*

Process wastewater applied	
4,704,622.00	*gallons*
173.26	*acre-inches*
2.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/06/2022: Wheat, silage, soft dough

Field name: 7 Crop: Wheat, silage, soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	131.74	16.44	195.20	1,530.96
Fresh water	20.42	0.00	0.00	777.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.16	16.44	195.20	2,308.28
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	119.11	31.16	218.84	858.72
Nutrient balance	40.05	-14.72	-23.63	1,449.56
Applied to removed ratio	1.34	0.53	0.89	2.69

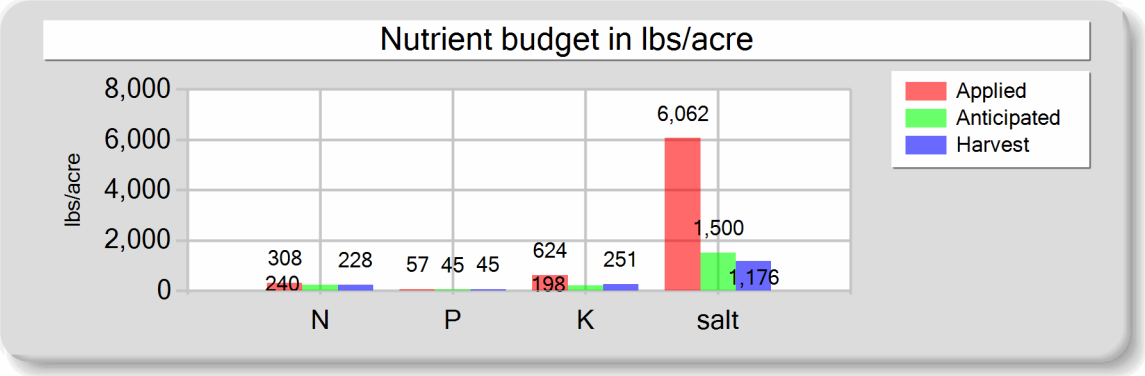
Fresh water applied	
6,315,156.00	*gallons*
232.57	*acre-inches*
5.81	*inches/acre*

Process wastewater applied	
859,896.00	*gallons*
31.67	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/11/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	259.18	57.32	624.16	3,931.60
Fresh water	41.79	0.00	0.00	2,130.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.97	57.32	624.16	6,061.79
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	228.33	45.29	250.97	1,175.60
Nutrient balance	79.64	12.03	373.19	4,886.19
Applied to removed ratio	1.35	1.27	2.49	5.16

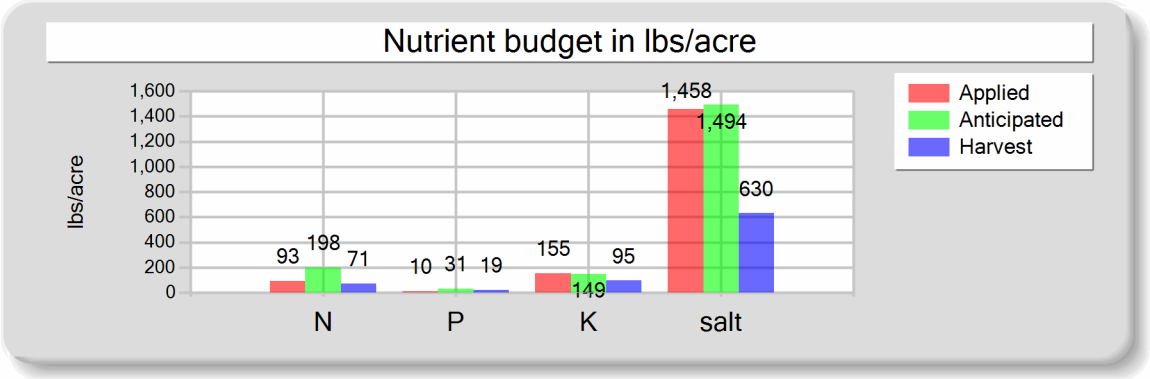
Fresh water applied	
43,118,964.00	*gallons*
1,587.92	*acre-inches*
39.70	*inches/acre*

Process wastewater applied	
3,511,136.00	*gallons*
129.30	*acre-inches*
3.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/04/2022: Wheat, silage, soft dough

Field name: 8 Crop: Wheat, silage, soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	84.27	9.84	155.27	1,352.35
Fresh water	1.71	0.00	0.00	106.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.98	9.84	155.27	1,458.39
Anticipated crop nutrient removal	198.00	30.60	149.40	1,494.00
Actual crop nutrient removal	70.59	19.40	95.38	630.48
Nutrient balance	22.39	-9.56	59.89	827.92
Applied to removed ratio	1.32	0.51	1.63	2.31

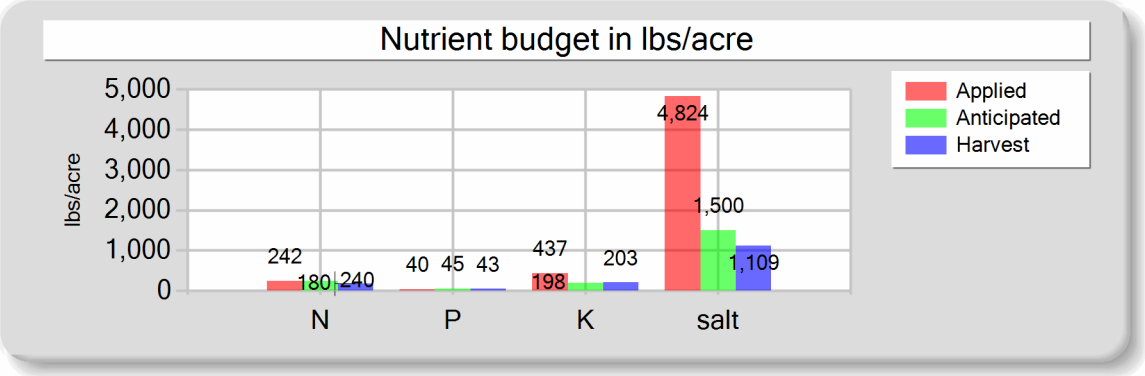
Fresh water applied	
2,470,923.00	*gallons*
91.00	*acre-inches*
2.60	*inches/acre*

Process wastewater applied	
525,178.00	*gallons*
19.34	*acre-inches*
0.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/07/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.69	40.32	437.14	2,697.65
Fresh water	50.22	0.00	0.00	2,126.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.91	40.32	437.14	4,823.82
Anticipated crop nutrient removal	240.00	45.00	198.00	1,500.00
Actual crop nutrient removal	179.73	43.13	203.09	1,108.92
Nutrient balance	62.18	-2.82	234.05	3,714.90
Applied to removed ratio	1.35	0.93	2.15	4.35

Fresh water applied	
39,065,876.00	*gallons*
1,438.66	*acre-inches*
41.10	*inches/acre*

Process wastewater applied	
2,135,732.00	*gallons*
78.65	*acre-inches*
2.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure

Sample and source description: Manure

Sample date: 11/28/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 35.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,000.00	8,000.00	26,500.00							
DL	100.00	200.00	200.00							

Dry Manure

Sample and source description: Dry Manure

Sample date: 05/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	8,600.00	17,200.00							
DL	100.00	200.00	200.00							

Dry Manure

Sample and source description: Dry Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 27.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,000.00	8,400.00	24,300.00							
DL	100.00	200.00	200.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon

Sample and source description: Lagoon

Sample date: 11/28/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	673.00	478.00	0.00	0.00	78.60	1,240.00								16,200.00	10,800
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	797.00	429.00	0.00	0.00	105.00	933.00								9,360.00	6,220
DL	2.00	10.00	5.00	5.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 05/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	630.00	449.00	0.00	0.00	107.00	1,050.00								4,820.00	3,200
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 08/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	263.00	255.00	0.00	0.00	68.80	787.00								9,150.00	6,080
DL	10.00	2.00	2.00	2.00	0.20	0.50								100.00	10

Lagoon

Sample and source description: Lagoon

Sample date: 11/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	638.00	631.00	0.00	0.00	69.60	793.00								9,120.00	6,060
DL	10.00	2.00	0.20	0.20	0.20	0.50								100.00	10

C. FRESH WATER ANALYSES

1

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	42.60	0.00	42.60	123.00	61.00	43.00	520.00	0.00	22.60	54.00	1,330.00	900
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

2

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	23.80	0.00	23.80	83.00	44.00	31.00	360.00	0.00	14.00	41.00	916.00	610
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

3

3

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.50	0.00	15.50	84.00	45.00	32.00	380.00	0.00	14.00	43.00	868.00	590
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

4

Irrigation Well

Sample description: Irrigation Well

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.40	0.00	15.40	78.00	40.00	34.00	370.00	0.00	13.50	47.00	865.00	560
DL	0.10	0.10	0.10	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

5

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	34.90	0.00	34.90	107.00	57.00	47.00	490.00	0.00	14.60	55.00	1,200.00	800
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

8a

8a

Irrigation Water

Sample description: Irrigation Water

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.90	0.00	2.90	18.00	10.00	17.00	90.00	0.00	3.80	12.00	237.00	180
DL	0.50	0.20	0.10	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

9

Irrigation Well

Sample description: Irrigation Well

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.20	0.00	13.20	61.00	29.00	28.00	300.00	0.00	9.50	19.00	655.00	430
DL	0.10	0.10	0.10	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

A

Domestic Well

Sample description: Domestic Well

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.70	0.00	6.70	31.00	15.00	18.00	170.00	0.00	5.00	13.00	377.00	270
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

B

B

Domestic Well

Sample description: Domestic Well

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.90	0.00	9.90	43.00	21.00	22.00	230.00	0.00	6.50	16.00	505.00	370
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

C

Domestic Well

Sample description: Domestic Well

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.90	0.00	9.90	36.00	18.00	22.00	170.00	0.00	4.00	15.00	417.00	320
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

D

Domestic Well

Sample description: Domestic Well

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	22.10	0.00	22.10	110.00	58.00	41.00	510.00	0.00	20.20	69.00	1,200.00	790
DL	0.20	0.20	0.20	1.00	1.00	1.00	10.00	10.00	0.17	1.00	1.00	20

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/29/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	5,500.00	41,700.00		11.90
DL	500.00	200.00	200.00		0.05

1 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,800.00	14,400.00		6.64
DL	500.00	200.00	200.00		0.05

10 - 11/13/2022: Wheat, silage, soft dough

10 - 11/13/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	3,900.00	20,000.00		12.80
DL	200.00	200.00	500.00		0.05

10 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,900.00	12,300.00		6.41
DL	500.00	200.00	200.00		0.05

11 - 11/04/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	5,600.00	33,800.00		12.60
DL	500.00	200.00	200.00		0.05

11 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,800.00	15,600.00		6.86
DL	500.00	200.00	200.00		0.05

12 - 11/06/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	3,000.00	15,100.00		8.99
DL	500.00	200.00	200.00		0.05

12 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,600.00	2,600.00	8,700.00		4.64
DL	500.00	200.00	200.00		0.05

13 - 11/07/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	4,800.00	36,800.00		12.30
DL	500.00	200.00	200.00		0.05

13 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	2,600.00	12,700.00		5.81
DL	500.00	200.00	200.00		0.05

14 - 11/05/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	3,000.00	15,800.00		6.63
DL	500.00	200.00	200.00		0.05

14 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	3,000.00	13,200.00		6.62
DL	500.00	200.00	200.00		0.05

2 - 10/29/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	4,700.00	30,500.00		10.70
DL	500.00	200.00	200.00		0.05

2 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,000.00	13,900.00		5.27
DL	500.00	200.00	200.00		0.05

3 - 10/30/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	5,300.00	42,300.00		12.60
DL	500.00	200.00	200.00		0.05

3 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,500.00	11,400.00		6.04
DL	500.00	200.00	200.00		0.05

4 - 10/31/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,300.00	3,200.00	14,600.00		8.71
DL	500.00	200.00	200.00		0.05

4 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,700.00	10,300.00		5.55
DL	500.00	200.00	200.00		0.05

5 - 10/31/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,900.00	23,700.00		10.10
DL	500.00	200.00	200.00		0.05

5 - 06/04/2023: Corn, silage

Corn

Sample and source description: Corn

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	3,000.00	13,300.00		6.01
DL	500.00	200.00	200.00		0.05

6 - 11/02/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	3,700.00	30,300.00		10.40
DL	500.00	200.00	200.00		0.05

6 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,800.00	11,900.00		5.70
DL	500.00	200.00	200.00		0.05

7 - 11/06/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	4,500.00	31,600.00		12.40
DL	500.00	200.00	200.00		0.05

7 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,400.00	13,300.00		6.23
DL	500.00	200.00	200.00		0.05

8 - 11/04/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	3,600.00	17,700.00		11.70
DL	500.00	200.00	200.00		0.05

8 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	2,400.00	11,300.00		6.17
DL	500.00	200.00	200.00		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

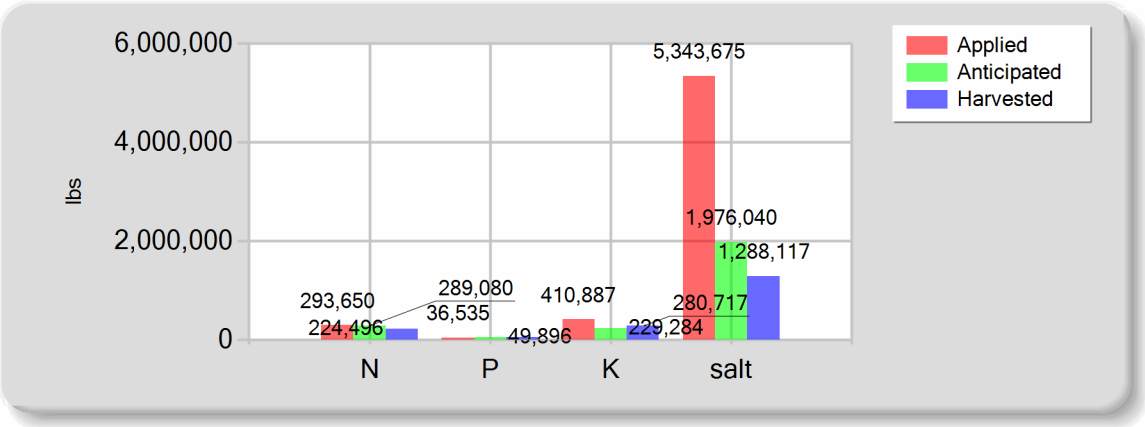
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

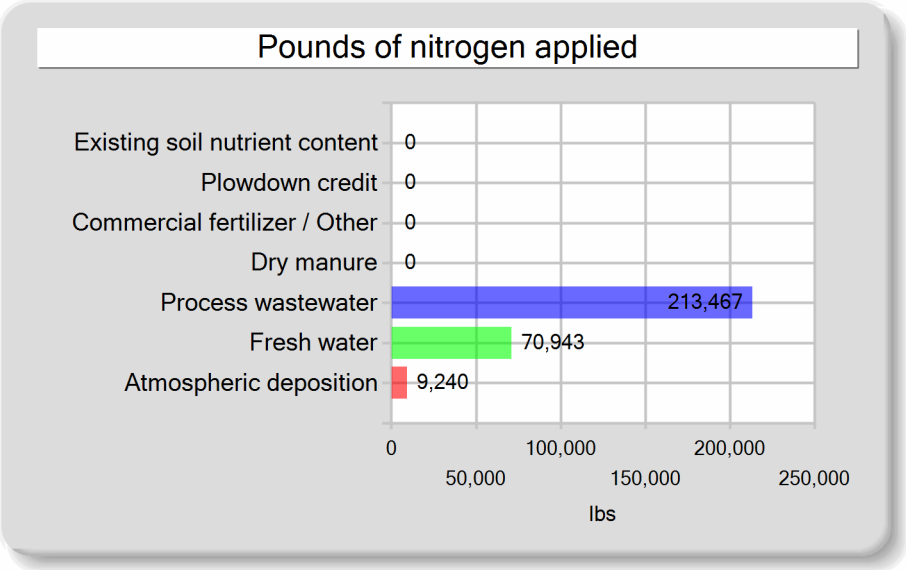
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

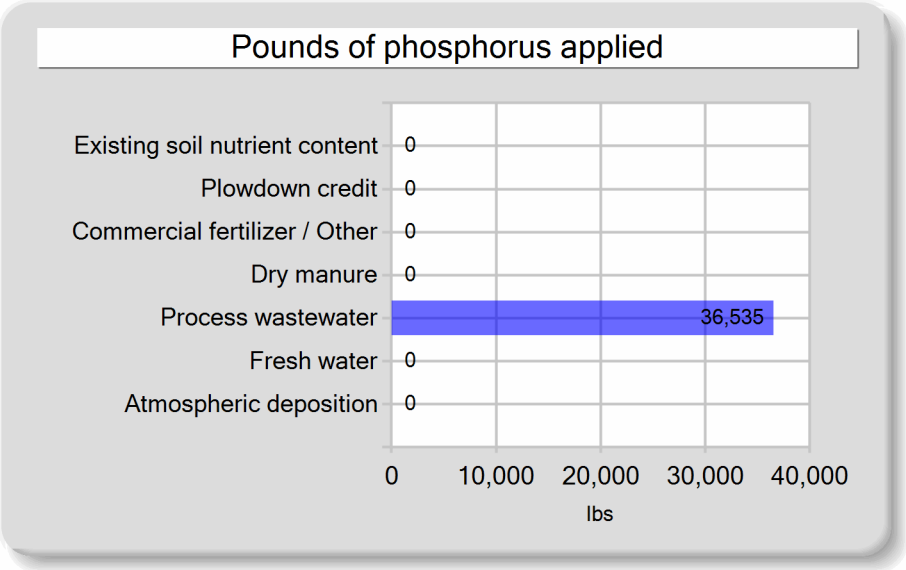
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213,467.18	36,535.00	410,887.05	2,542,274.51
Fresh water	70,942.64	0.00	0.00	2,801,400.56
Atmospheric deposition	9,240.00	0.00	0.00	0.00
Total nutrients applied	293,649.82	36,535.00	410,887.05	5,343,675.07
Anticipated crop nutrient removal	289,080.00	49,896.00	229,284.00	1,976,040.00
Actual crop nutrient removal	224,495.88	54,403.31	280,716.77	1,288,117.36
Nutrient balance	69,153.94	-17,868.31	130,170.28	4,055,557.71
Applied to removed ratio	1.31	0.67	1.46	4.15

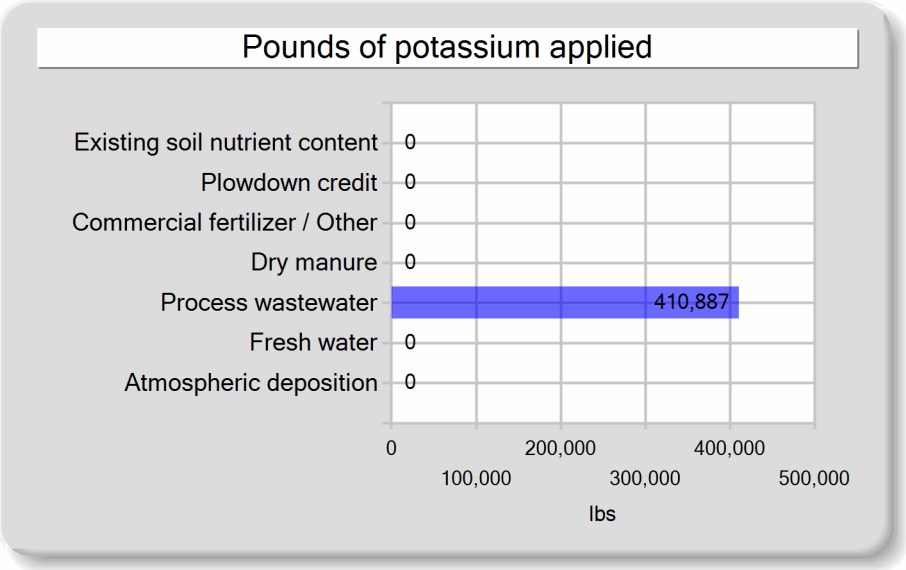
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

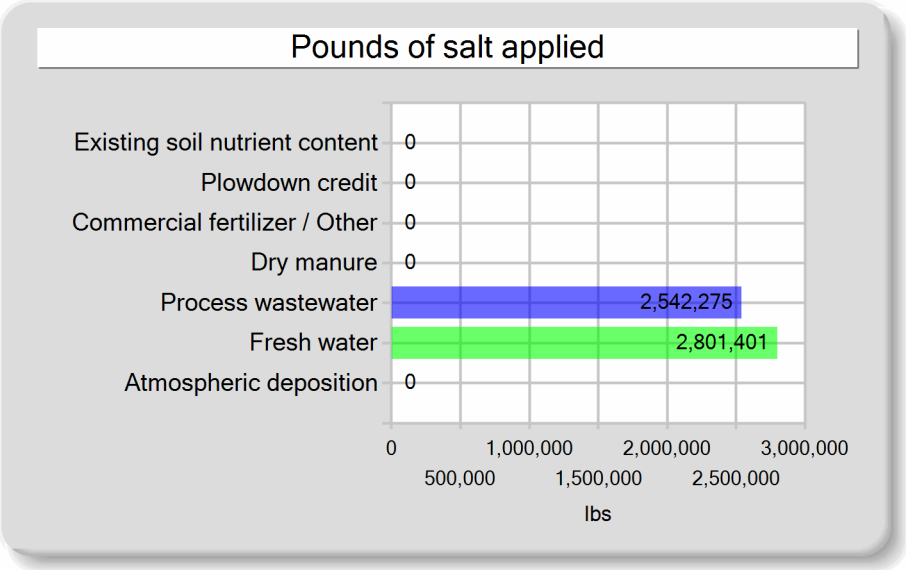


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Well 8 was out of service in 2023.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ed Nunes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
5-30-24	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Lab No. : **VI 2346141**
Customer No. : **4019696**
Reference : **3151**

Laboratory Report

Introduction: This report package contains a total of 15 pages divided into 3 sections:

Case Narrative	(2 pages)	: An overview of the work performed at FGL.
Sample Results	(8 pages)	: Results for each sample submitted.
Quality Control	(5 pages)	: Supporting Quality Control (QC) results.

Case Narrative

This Case Narrative pertains to the following samples:

Sample Description	Date Sampled	Date Received	FGL Lab No.	Matrix
A	09/11/2023	09/11/2023	VI 2346141-001	DW
B	09/11/2023	09/11/2023	VI 2346141-002	DW
C	09/11/2023	09/11/2023	VI 2346141-003	DW
D	09/11/2023	09/11/2023	VI 2346141-004	DW
1	09/11/2023	09/11/2023	VI 2346141-005	AGW
2	09/11/2023	09/11/2023	VI 2346141-006	AGW
3	09/11/2023	09/11/2023	VI 2346141-007	AGW
4	09/11/2023	09/11/2023	VI 2346141-008	AGW

Sampling and Receipt Information:

All samples were received in acceptable condition and within temperature requirements, unless noted on the Condition Upon Receipt (CUR) form. All samples were received, prepared and analyzed within the method specified holding times except those as listed in the table below. All samples arrived on ice. All samples were checked for pH if acid or base preservation is required (except for VOAs). For details of sample receipt information, please see the associated Chain of Custody and Condition Upon Receipt Form.

Samples Over Hold Time

Lab No	Analyte Method	Maximum Hold Time	Actual Hold Time
VI 2346141-006	Chloride	28 days	30.3 days
VI 2346141-006	Nitrate Nitrogen	48 hours	727.7 hours
VI 2346141-006	Sulfate Sulfur	28 days	30.3 days

Quality Control: All samples were prepared and analyzed according to established quality control criteria. Any exceptions are noted in the Quality Control Section of this report.

Test Summary	
EPA 200.7	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
EPA 300.0	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
EPA 351.2	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 2540 C	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

SM 4500-H+B	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-NO3 F	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Certification: I certify that this data package is in compliance with ELAP standards, both technically and for completeness, except for any conditions listed above and in the QC Section. Release of the data contained in this data package is authorized by the Laboratory Director or his designee, as verified by the following electronic signature. This report shall not be reproduced except in full, without the written approval of the laboratory.

KD: JRD

Approved By **Kelly A. Dunnahoo, B.S.** Digitally signed by Kelly A. Dunnahoo, B.S.
Title: Laboratory Director
Date: 2023-10-16

Page 2 of 15

October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : A
Project : Ed Nunes Dairy

Lab No. : VI 2346141-001
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 08:05
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	140	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	21:42	amm
Bicarbonate	170	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	21:42	amm
Carbonate	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	21:42	amm
Hydroxide	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	21:42	amm
Chloride	13	1	mg/L	500[2]	1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:52	ldm
Nitrate Nitrogen	6.7	0.1	mg/L	10	1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:52	ldm
Conductivity	377	1	umhos/cm	1600[2]	1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	21:42	amm
Sulfate Sulfur	5.00	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:52	ldm
Solids, Total Dissolved (TDS)	270	20	mg/L	1000[2]	1		09/14/2023	10:30	ctl	SM 2540 C	09/15/2023	11:00	ctl
Calcium	31	1	mg/L		1	l	09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:10	ac
Magnesium	15	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:10	ac
Potassium	5	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:10	ac
Sodium	18	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:10	ac

DQF Flags Definition:
U Constituent results were non-detect.
I The RPD for the laboratory duplicate exceeded laboratory criteria.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

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TEL: (805)392-2000
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Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
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563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
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Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
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Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : B
Project : Ed Nunes Dairy

Lab No. : VI 2346141-002
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 07:55
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	190	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:02	amm
Bicarbonate	230	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:02	amm
Carbonate	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:02	amm
Hydroxide	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:02	amm
Chloride	16	1	mg/L	500[2]	1		09/12/2023	14:47	ldm	EPA 300.0	09/13/2023	01:32	ldm
Nitrate Nitrogen	9.9	0.1	mg/L	10	1		09/12/2023	14:47	ldm	EPA 300.0	09/13/2023	01:32	ldm
Conductivity	505	1	umhos/cm	1600[2]	1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:02	amm
Sulfate Sulfur	6.50	0.17	mg/L		1		09/12/2023	14:47	ldm	EPA 300.0	09/13/2023	01:32	ldm
Solids, Total Dissolved (TDS)	370	20	mg/L	1000[2]	1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	43	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:54	ac
Magnesium	21	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:54	ac
Potassium	6	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:54	ac
Sodium	22	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:54	ac

DQF Flags Definition:
U Constituent results were non-detect.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

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2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
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563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
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San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
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Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : C
Project : Ed Nunes Dairy

Lab No. : VI 2346141-003
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 07:40
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	140	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	03:53	amm
Bicarbonate	170	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	03:53	amm
Carbonate	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	03:53	amm
Hydroxide	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	03:53	amm
Chloride	15	1	mg/L	500[2]	1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	21:21	ldm
Nitrate Nitrogen	9.9	0.1	mg/L	10	1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	21:21	ldm
Conductivity	417	1	umhos/cm	1600[2]	1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	03:53	amm
Sulfate Sulfur	4.00	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	21:21	ldm
Solids, Total Dissolved (TDS)	320	20	mg/L	1000[2]	1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	36	1	mg/L		1	l	09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:19	ac
Magnesium	18	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:19	ac
Potassium	6	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:19	ac
Sodium	22	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:19	ac

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

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Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
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Chico, CA 95926
TEL: (530)343-5818
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9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : D
Project : Ed Nunes Dairy

Lab No. : VI 2346141-004
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 07:35
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Drinking Water

Sample Results - Inorganic

Constituent	Result	RL	Units	MCL/AL	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	420	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:23	amm
Bicarbonate	510	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:23	amm
Carbonate	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:23	amm
Hydroxide	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:23	amm
Chloride	69	1	mg/L	500[2]	1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	19:15	ldm
Nitrate Nitrogen	22.1	0.2*	mg/L	10	2		09/12/2023	10:31	ldm	EPA 300.0	09/13/2023	04:39	ldm
Conductivity	1200	1	umhos/cm	1600[2]	1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:23	amm
Sulfate Sulfur	20.2	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	19:15	ldm
Solids, Total Dissolved (TDS)	790	20	mg/L	1000[2]	1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	110	1	mg/L		1	l	09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:23	ac
Magnesium	58	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:23	ac
Potassium	6	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:23	ac
Sodium	41	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	17:23	ac

DQF Flags Definition:
U Constituent results were non-detect.
I The RPD for the laboratory duplicate exceeded laboratory criteria.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution
MCL = Maximum Contamination Level. 2 - Secondary Standard. 3 - CDPH Notification Level. AL = Regulatory Action Level.

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Stockton, CA 95215
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FAX: (209)942-0423
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Chico, CA 95926
TEL: (530)343-5818
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San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
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Visalia, CA 93291
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FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 1
Project : Ed Nunes Dairy

Lab No. : VI 2346141-005
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 07:10
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	430	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	03:07	amm
Bicarbonate	520	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	03:07	amm
Carbonate	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	03:07	amm
Hydroxide	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	03:07	amm
Chloride	54	1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	18:54	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:12	lcr
Nitrate Nitrogen	42.6	0.5*	mg/L		5		09/12/2023	10:31	ldm	EPA 300.0	09/13/2023	03:37	ldm
Nitrogen, Total as Nitrogen	24.0	0.5	mg/L		1	l	09/21/2023	09:23	sta	Calc.	09/25/2023	20:12	lcr
Nitrate + Nitrite as N	42.6	0.5*	mg/L		5		09/12/2023	10:31	ldm	EPA 300.0	09/13/2023	03:37	ldm
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:12	lcr
Conductivity	1330	1	umhos/cm		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	03:07	amm
Sulfate Sulfur	22.6	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	18:54	ldm
Solids, Total Dissolved (TDS)	900	20	mg/L		1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	123	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	20:07	ac
Magnesium	61	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	20:07	ac
Sodium	43	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	20:07	ac

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

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Stockton, CA 95215
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Chico, CA 95926
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 2
Project : Ed Nunes Dairy

Lab No. : VI 2346141-006
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 07:18
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	300	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:23	amm
Bicarbonate	360	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:23	amm
Carbonate	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:23	amm
Hydroxide	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:23	amm
Chloride	41	1	mg/L		1	T	10/11/2023	14:58	ldm	EPA 300.0	10/11/2023	18:32	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:15	lcr
Nitrate Nitrogen	22.0	0.3*	mg/L		3	T	10/11/2023	14:58	ldm	EPA 300.0	10/12/2023	01:51	ldm
Nitrogen, Total as Nitrogen	23.8	0.5	mg/L		1	l	09/21/2023	09:23	sta	Calc.	09/25/2023	20:15	lcr
Nitrate + Nitrite as N	23.8	0.4	mg/L		1	T	10/11/2023	11:50	lfs	SM 4500-NO3 F	10/11/2023	16:01	lfs
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:15	lcr
Conductivity	916	1	umhos/cm		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:23	amm
Sulfate Sulfur	14.0	0.17	mg/L		1	T	10/11/2023	14:58	ldm	EPA 300.0	10/11/2023	18:32	ldm
Solids, Total Dissolved (TDS)	610	20	mg/L		1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	83	1	mg/L		1	l	09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	16:18	ac
Magnesium	44	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	16:18	ac
Sodium	31	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	16:18	ac

DQF Flags Definition:
U Constituent results were non-detect.
T Exceeded method/regulatory-specific holding time.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

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853 Corporation Street
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Stockton, CA 95215
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 3
Project : Ed Nunes Dairy

Lab No. : VI 2346141-007
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 06:40
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	310	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	22:10	amm
Bicarbonate	380	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	22:10	amm
Carbonate	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	22:10	amm
Hydroxide	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	22:10	amm
Chloride	43	1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	16:49	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:17	lcr
Nitrate Nitrogen	15.5	0.1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	16:49	ldm
Nitrogen, Total as Nitrogen	15.5	0.5	mg/L		1	l	09/21/2023	09:23	sta	Calc.	09/25/2023	20:17	lcr
Nitrate + Nitrite as N	15.5	0.1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	16:49	ldm
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:17	lcr
Conductivity	868	1	umhos/cm		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	22:10	amm
Sulfate Sulfur	14.0	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	16:49	ldm
Solids, Total Dissolved (TDS)	590	20	mg/L		1		09/13/2023	10:20	ctl	SM 2540 C	09/14/2023	11:40	ctl
Calcium	84	1	mg/L		1	l	09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:45	ac
Magnesium	45	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:45	ac
Sodium	32	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	15:45	ac

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 4
Project : Ed Nunes Dairy

Lab No. : VI 2346141-008
Customer No. : 4019696
Reference : 3151
Sampled On : September 11, 2023 at 06:45
Sampled By : Darin
Received On : September 11, 2023 at 15:20
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	300	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:47	amm
Bicarbonate	370	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:47	amm
Carbonate	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:47	amm
Hydroxide	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:47	amm
Chloride	47	1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:10	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:19	lcr
Nitrate Nitrogen	15.4	0.1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:10	ldm
Nitrogen, Total as Nitrogen	15.4	0.5	mg/L		1	l	09/21/2023	09:23	sta	Calc.	09/25/2023	20:19	lcr
Nitrate + Nitrite as N	15.4	0.1	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:10	ldm
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/21/2023	09:23	sta	EPA 351.2	09/25/2023	20:19	lcr
Conductivity	865	1	umhos/cm		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	18:47	amm
Sulfate Sulfur	13.5	0.17	mg/L		1		09/12/2023	10:31	ldm	EPA 300.0	09/12/2023	17:10	ldm
Solids, Total Dissolved (TDS)	560	20	mg/L		1		09/14/2023	10:30	ctl	SM 2540 C	09/15/2023	11:00	ctl
Calcium	78	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:01	ac
Magnesium	40	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:01	ac
Sodium	34	1	mg/L		1		09/14/2023	06:45	ejc	EPA 200.7	09/14/2023	19:01	ac

DQF Flags Definition:
U Constituent results were non-detect.
I The RPD for the laboratory duplicate exceeded laboratory criteria.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

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October 16, 2023
Sentry Ag Service

Lab No. : VI 2346141
Customer No. : 4019696

Quality Control - Metals

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Metals								
Calcium	200.7	09/14/2023:210310EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210309EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	93.4%	85-115	
	200.7	09/14/2023:210310EJC	LCS	mg/L	12.00	97.3%	85-115	
	200.7	09/14/2023:210309EJC	MS	mg/L	12.00	43.3%	<¼	406
	200.7	09/14/2023:210310EJC	MS	mg/L	12.00	105%	75-125	
		(STK2352400-005)	MSD	mg/L	12.00	71.3%	75-125	435
	200.7	(SP 2315438-001)	MSD	mg/L	12.00	61.1%	<1/4	
			MSRPD	mg/L		2.0%	≤20.0	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		17.6%	≤20.0	
			MS	mg/L	12.00	107%	75-125	
	200.7	09/14/2023:210309EJC	MS	mg/L	12.00	99.5%	75-125	
		(SP 2315435-001)	MSD	mg/L	12.00	108%	75-125	
	200.7	(STK2352400-002)	MSD	mg/L	12.00	94.4%	75-125	
			MSRPD	mg/L		6.4%	≤20.0	
	200.7	09/14/2023:210309EJC	MSRPD	mg/L		1.0%	≤20.0	
Magnesium	200.7	09/14/2023:210309EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210310EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210309EJC	LCS	mg/L	12.00	98.3%	85-115	
	200.7	09/14/2023:210310EJC	LCS	mg/L	12.00	102%	85-115	
			MS	mg/L	12.00	106%	75-125	
	200.7	09/14/2023:210309EJC	MS	mg/L	12.00	86.7%	75-125	
	200.7	(STK2352400-005)	MSD	mg/L	12.00	1.63%	<1/4	
	200.7	(SP 2315438-001)	MSD	mg/L	12.00	90.5%	75-125	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		17.5%	≤20	
	200.7	09/14/2023:210309EJC	MSRPD	mg/L		1.3%	≤20	
			MS	mg/L	12.00	99.1%	75-125	
	200.7	09/14/2023:210310EJC	MS	mg/L	12.00	113%	75-125	
		(STK2352400-002)	MSD	mg/L	12.00	72.6%	<1/4	
	200.7	(SP 2315435-001)	MSD	mg/L	12.00	104%	75-125	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		5.0%	≤20	
	200.7	09/14/2023:210309EJC	MSRPD	mg/L		1.6%	≤20	
Potassium	200.7	09/14/2023:210310EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210309EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210310EJC	LCS	mg/L	12.00	97.1%	85-115	
	200.7	09/14/2023:210309EJC	LCS	mg/L	12.00	93.4%	85-115	
			MS	mg/L	12.00	98.7%	75-125	
	200.7	09/14/2023:210310EJC	MS	mg/L	12.00	104%	75-125	
		(STK2352400-005)	MSD	mg/L	12.00	75.0%	75-125	
	200.7	(SP 2315438-001)	MSD	mg/L	12.00	99.9%	75-125	
			MSRPD	mg/L		1.0%	≤20.0	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		17.6%	≤20.0	
			MS	mg/L	12.00	108%	75-125	
	200.7	09/14/2023:210309EJC	MS	mg/L	12.00	95.8%	75-125	
	200.7	(STK2352400-002)	MSD	mg/L	12.00	95.7%	75-125	
	200.7	(SP 2315435-001)	MSD	mg/L	12.00	101%	75-125	

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October 16, 2023

Sentry Ag Service

Lab No. : VI 2346141
Customer No. : 4019696

Quality Control - Metals

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
			MSRPD	mg/L		2.0%	≤20.0	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		6.0%	≤20.0	
Sodium	200.7	09/14/2023:210309EJC	Blank	mg/L		ND	<1	
	200.7	09/14/2023:210310EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	97.8%	85-115	
	200.7	09/14/2023:210309EJC	LCS	mg/L	12.00	94.7%	85-115	
	200.7	09/14/2023:210310EJC	MS	mg/L	12.00	186%	<¼	406
	200.7	09/14/2023:210309EJC	MS	mg/L	12.00	80.8%	75-125	
		(SP 2315438-001)	MSD	mg/L	12.00	85.5%	75-125	
	200.7	(STK2352400-005)	MSD	mg/L	12.00	-167%	<1/4	
			MSRPD	mg/L		18.5%	≤20.0	
	200.7	09/14/2023:210309EJC	MSRPD	mg/L		1.4%	≤20.0	
			MS	mg/L	12.00	89.3%	75-125	
	200.7	09/14/2023:210310EJC	MS	mg/L	12.00	200%	<¼	406
		(STK2352400-002)	MSD	mg/L	12.00	86.0%	75-125	
	200.7	(SP 2315435-001)	MSD	mg/L	12.00	104%	75-125	
			MSRPD	mg/L		2.5%	≤20.0	
	200.7	09/14/2023:210310EJC	MSRPD	mg/L		3.4%	≤20.0	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

DQO : Data Quality Objective - This is the criteria against which the quality control data is compared.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Explanation

406 : Matrix Spike (MS) not within the Acceptance Range (AR) because of high analyte concentration in the sample. Data was accepted based on the LCS or CCV recovery.

435 : Sample matrix may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

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October 16, 2023
Sentry Ag Service

Lab No. : VI 2346141
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Wet Chem								
Alkalinity (as CaCO3)	2320B	09/18/2023:210413AMM	ND	mg/L		0.4%	10	435
	2320B	(VI 2346309-001)	Dup	mg/L		6.26%	10	
		(SP 2315465-001)	Dup	mg/L		0.1%	10	
		(SP 2315663-001)	Dup	mg/L		0.4%	10	
Bicarbonate	2320B	(CH 2377803-001)	Dup	mg/L		0.4%	10	
	2320B	(VI 2346309-001)	Dup	mg/L		8.88%	10	
		(SP 2315465-001)	Dup	mg/L		0.1%	10	
		(SP 2315663-001)	Dup	mg/L		0.3%	10	
Carbonate	2320B	(CH 2377803-001)	Dup	mg/L			10	
	2320B	(VI 2346309-001)	Dup	mg/L		15.4%	10	440
E. C.	2320B	(CH 2377803-001)	Dup	umhos/cm		0.6%	5	
	2320B	(VI 2346309-001)	Dup	umhos/cm		0%	5	
		(SP 2315465-001)	Dup	umhos/cm		0.2%	5	
		(SP 2315663-001)	Dup	umhos/cm		0%	5	
Solids, Total Dissolved	2540CE	09/13/2023:210246CTL	Blank	mg/L		ND	<20	
			LCS	mg/L	991.5	101%	90-110	
		(VI 2346141-005)	Dup	mg/L		1.75%	5	
		(VI 2346141-005)	Dup	mg/L		2.07%	5	
	2540CE	09/14/2023:210327CTL	Blank	mg/L		ND	<20	
			LCS	mg/L	991.5	101%	90-110	
		(CC 2383073-001)	Dup	mg/L		1.58%	5	
		(CC 2383073-001)	Dup	mg/L		0.6%	5	
Chloride	300.0	09/12/2023:210259LDM	Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	103 %	90-110	
			MS	mg/L	50.00	98.6 %	67-117	
		(CH 2377560-001)	MSD	mg/L	50.00	98.7 %	67-117	
			MSRPD	mg/L	10.00	0.05%	≤7	
			MS	mg/L	50.00	101 %	67-117	
		(CH 2377560-002)	MSD	mg/L	50.00	101 %	67-117	
			MSRPD	mg/L	10.00	0.07%	≤7	
			Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	103 %	90-110	
			MS	mg/L	50.00	103 %	67-117	
		(STK2352408-001)	MSD	mg/L	50.00	103 %	67-117	
			MSRPD	mg/L	10.00	0.1%	≤7	
			MS	mg/L	50.00	100 %	67-117	
		(VI 2346141-002)	MSD	mg/L	50.00	100 %	67-117	
			MSRPD	mg/L	10.00	0.3%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	99.3%	90-110	
			MS	mg/L	50.00	92.5%	67-117	
		(CH 2378529-001)	MSD	mg/L	50.00	92.5%	67-117	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	50.00	91.0%	67-117	
		(CH 2378530-001)	MSD	mg/L	50.00	91.5%	67-117	
			MSRPD	mg/L		0.4%	≤7	
Nitrate + Nitrite as N	300.0	09/12/2023:210259LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	104 %	90-110	
			MS	mg/L	40.00	100 %	86-112	
		(CH 2377560-001)	MSD	mg/L	40.00	100 %	86-112	

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October 16, 2023
Sentry Ag Service

Lab No. : VI 2346141
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
			MSRPD	mg/L	10.00	0.07%	≤7	
			MS	mg/L	40.00	103 %	86-112	
		(CH 2377560-002)	MSD	mg/L	40.00	103 %	86-112	
			MSRPD	mg/L	10.00	0.1%	≤7	
Nitrate Nitrogen	300.0	09/12/2023:210259LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	104 %	90-110	
			MS	mg/L	40.00	100 %	86-112	
		(CH 2377560-001)	MSD	mg/L	40.00	100 %	86-112	
			MSRPD	mg/L	10.00	0.07%	≤7	
			MS	mg/L	40.00	103 %	86-112	
		(CH 2377560-002)	MSD	mg/L	40.00	103 %	86-112	
			MSRPD	mg/L	10.00	0.1%	≤7	
			Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	104 %	90-110	
			MS	mg/L	40.00	105 %	86-112	
		(STK2352408-001)	MSD	mg/L	40.00	105 %	86-112	
			MSRPD	mg/L	10.00	0.06%	≤7	
			MS	mg/L	40.00	89.1 %	86-112	
		(VI 2346141-002)	MSD	mg/L	40.00	88.6 %	86-112	
			MSRPD	mg/L	10.00	0.3%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	99.5%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(CH 2378529-001)	MSD	mg/L	40.00	103%	86-112	
			MSRPD	mg/L		0.2%	≤7	
			MS	mg/L	40.00	102%	86-112	
		(CH 2378530-001)	MSD	mg/L	40.00	102%	86-112	
			MSRPD	mg/L		0.2%	≤7	
Sulfate Sulfur	300.0	09/12/2023:210259LDM	Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	104 %	90-110	
			MS	mg/L	100.0	99.7 %	18-165	
		(CH 2377560-001)	MSD	mg/L	100.0	99.9 %	18-165	
			MSRPD	mg/L	10.00	0.2%	≤7	
			MS	mg/L	100.0	103 %	18-165	
		(CH 2377560-002)	MSD	mg/L	100.0	103 %	18-165	
			MSRPD	mg/L	10.00	0.02%	≤7	
			Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	104 %	90-110	
			MS	mg/L	100.0	104 %	18-165	
		(STK2352408-001)	MSD	mg/L	100.0	103 %	18-165	
			MSRPD	mg/L	10.00	0.1%	≤7	
			MS	mg/L	100.0	103 %	18-165	
		(VI 2346141-002)	MSD	mg/L	100.0	103 %	18-165	
			MSRPD	mg/L	10.00	0.3%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	99.6%	90-110	
			MS	mg/L	100.0	102%	18-165	
		(CH 2378529-001)	MSD	mg/L	100.0	102%	18-165	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	100.0	99.2%	18-165	
		(CH 2378530-001)	MSD	mg/L	100.0	99.6%	18-165	

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October 16, 2023

Sentry Ag Service

Lab No. : VI 2346141
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
			MSRPD	mg/L		0.3%	≤7	
Nitrogen, Total Kjeldahl	351.2	09/21/2023:210595STA	Blank	mg/L		ND	<0.5	
			LCS	mg/L	12.00	91.7%	73-124	
			MS	mg/L	12.00	89.3%	90-110	435
		(STK2352400-001)	MSD	mg/L	12.00	88.1%	90-110	435
			MSRPD	mg/L		1.2%	≤20	
			MS	mg/L	12.00	89.7%	90-110	435
		(STK2352400-004)	MSD	mg/L	12.00	93.6%	90-110	
			MSRPD	mg/L		3.8%	≤20	
Nitrate + Nitrite as N	4500NO3F	10/11/2023:211472LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	100%	80-120	
			MS	mg/L	5.609	98.0%	66-125	
		(CH 2378768-001)	MSD	mg/L	5.609	99.3%	66-125	
			MSRPD	mg/L		1.2%	≤30.4	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

Dup : Duplicate Sample - A random sample with each batch is prepared and analyzed in duplicate. The relative percent difference is an indication of precision for the preparation and analysis.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Explanation

435 : Sample matrix may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

440 : Sample nonhomogeneity may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

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Laboratory Analysis Work Order

3151

SITE NAME: Ed Nunes Dairy 2846141 **LABORATORY:** VT | FGL 4-19696

Billing: Sentry Ag Services, LLC
P.O. Box 7750, Visalia, CA 93290

Authorized Copy Release to:
labs@sentryagservices.com

ANALYSIS TO BE COMPLETED

Irrigation/Ground Water (ELAP Standards)

- W1 EC, NO_3N (Dom)
- W2 EC, NO_3N, TDS, TN (Irr)
- W3 NH_4-N (Ammonium)
- W4 EC, NO_3N, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl, TDS (Dom, GM)
- W5 EC, NO_3N, TDS, TN, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl (Irr, GM)
- W6 NO_3N, NO_2 (Dom ILRP, Annually)
- W7 Ca, Mg, Na, K, HCO_3, CO_3, SO_4, Cl + Lab Filtering (GWM)
- W8 Other: _____

Pol 4.1c

Plant Tissue

- P1 TN, NO_3N, PO_4P, K (Mid Season - Wheat)
- P2 TN, P, K (Mid-season - Corn)
- P3 TN, TP, TK, Ash, %M (At Harvest)
- P4 TN, %M
- P5 % Moisture
- P6 NIR
- P7 Other: _____

Process Waste Water (lagoon)

- L1 EC, NH_4N, TKN, TP, TK, TDS (Quarterly)
- L2 EC, NO_3N, NH_4N, TKN, TP, TK, TDS, pH (Annually)
- L3 Ca, Mg, Na, HCO_3,CO_3, SO_4S, Cl (Biennially)
- L4 Other: _____

Manure

- M1 TN, TP, TK, %M (2/year)
- M2 TN, TP, K, %M, Ca, Mg, Na, S, Cl, ash (Biennially)
- M3 Other: _____

Soil

- S1 SP%, pH, EC, Ca, Mg, Na, K, ESP, LP, B, NO_3N, PO_4P, K-AA, Zn, Mn, Fe, Cu, SO_4S
- S2 S1 + CEC, CaCO3, OM, C:N, TN
- S3 NO_3N, NH_4N
- S4 Other: _____

	Sample ID	Description	Analysis	Date/Time	Sampled by	SAS USE ONLY: FIELD TESTS NH_3N *	pH	Temp
1	A	domestic well	W4	9/11/23 8:05am	Paul	0		
2	B			7:55am		0		
3	C			7:40am		0		
4	D			7:35am		0		
5	1	Irrigation well	W5	7:10am		0		
6	2			7:18am		0		
7	3			6:40am		0		
8	4			6:45am		0		
9								
10								
11								
12								

* Field Test of ammonium nitrogen may only be made by a trained technician. Positive test to be analyzed for ammonium nitrogen by the laboratory.

All samples are to follow the procedures noted in the Sampling & Analysis Plan of the NMP and the RWQCB specifications. Any samples taken outside of these procedures shall provide the procedures on the notes below. Additionally, if any preservatives are used in the collections or processing of sampes, please note below.

NOTES:

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	[signature]	SAS		9/11/23 3:20pm
2nd	[signature]	FGL	9/11/23 1520	
3rd	[signature]			9/11/23 1730
4th	[signature]		9/11/23 1730	

LABORATORY USE ONLY
Logged In By: _____ **Total Samples:** _____ **Laboratory No.:** _____

GLS 9/12/23
MC 1217

Inter-Laboratory Condition Upon Receipt (Attach to COC)

Sample Receipt at: STK CC CH VI

1. Number of ice chests/packages received: 1 Shipping tracking # GTC
2. Were samples received in a chilled condition? Temps: 2.1 / 4.6 / ___ / ___ / ___
   Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.

3. Do the number of bottles received agree with the COC?	Yes	No	N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5. VOAs checked for Headspace?	Yes	No	N/A
6. Were sample custody seals intact?	Yes	No	N/A
7. If required, was sample split for pH analysis?	Yes	No	N/A
8. Were all analyses within holding times at time of receipt?	Yes	No	
9. Verify sample date, time and sampler name	Yes	No	

Sign and date the COC, place in a ziplock and put in the same ice chest as the samples.
Sample Receipt Review completed by (initials): DA 7/14/07

Sample Receipt at SP:

1. Were samples received in a chilled condition? Temps: 2.1 / ___ / ___ / ___ / ___
   Acceptable is above freezing to 6° C. If many packages are received at one time check for tests/H.T.'s/rushes/
2. Shipping tracking numbers: 566100233 249 255

3. Do the number of bottles received agree with the COC?	Yes	No	N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.)	Yes	No	
5. Were sample custody seals intact?	Yes	No	N/A

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1. Were all requested analyses understood and acceptable?	Yes	No		
2. Did bottle labels correspond with the client's ID's?	Yes	No		
3. Were all bottles requiring sample preservation properly preserved? [Exception: Oil & Grease, VOA and CrVI verified in lab]	Yes	No	N/A	FGL
4. VOAs checked for Headspace?	Yes	No	N/A	
5. Have rush or project due dates been checked and accepted?	Yes	No	N/A	
6. Were all analyses within holding times at time of receipt?	Yes	No		

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): MX

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:

2. Person Contacted: _____ Phone Number: _____
   Initiated By: _____
   Problem:
   Resolution:

(Please use the back of this sheet for additional c
contacts)

(4019696)
Sentry Ag Service
VI 2346141
cda 09/12/2023 07:49:47
VI 2346141

October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Lab No. : **VI 2346219**
Customer No. : **4019696**
Reference : **3155**

Laboratory Report

Introduction: This report package contains a total of 9 pages divided into 3 sections:

Case Narrative	(1 page)	: An overview of the work performed at FGL.
Sample Results	(3 pages)	: Results for each sample submitted.
Quality Control	(5 pages)	: Supporting Quality Control (QC) results.

Case Narrative

This Case Narrative pertains to the following samples:

Sample Description	Date Sampled	Date Received	FGL Lab No.	Matrix
5	09/13/2023	09/13/2023	VI 2346219-001	AGW
8	09/13/2023	09/13/2023	VI 2346219-002	AGW
9	09/13/2023	09/13/2023	VI 2346219-003	AGW

Sampling and Receipt Information:

All samples were received in acceptable condition and within temperature requirements, unless noted on the Condition Upon Receipt (CUR) form. All samples were received, prepared and analyzed within the method specified holding times. All samples arrived on ice. All samples were checked for pH if acid or base preservation is required (except for VOAs). For details of sample receipt information, please see the associated Chain of Custody and Condition Upon Receipt Form.

Quality Control: All samples were prepared and analyzed according to established quality control criteria. Any exceptions are noted in the Quality Control Section of this report.

Test Summary	
EPA 200.7	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
EPA 300.0	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
EPA 351.2	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 2540 C	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-H+B	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)
SM 4500-NO3 F	Preparation and analysis performed by FGL-Santa Paula (FGL-SP ELAP# 1573)

Certification: I certify that this data package is in compliance with ELAP standards, both technically and for completeness, except for any conditions listed above and in the QC Section. Release of the data contained in this data package is authorized by the Laboratory Director or his designee, as verified by the following electronic signature. This report shall not be reproduced except in full, without the written approval of the laboratory.

KD: JRD

Approved By **Kelly A. Dunnahoo, B.S.** Digitally signed by Kelly A. Dunnahoo, B.S.
Title: Laboratory Director
Date: 2023-10-16

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 5
Project : Ed Nunes Dairy

Lab No. : VI 2346219-001
Customer No. : 4019696
Reference : 3155
Sampled On : September 13, 2023 at 06:38
Sampled By : Darin
Received On : September 13, 2023 at 14:42
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	400	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:11	amm
Bicarbonate	490	10	mg/L		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:11	amm
Carbonate	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:11	amm
Hydroxide	ND	10	mg/L		1	U	09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:11	amm
Chloride	55	1	mg/L		1		09/14/2023	11:22	ldm	EPA 300.0	09/15/2023	02:48	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:33	lcr
Nitrate Nitrogen	34.9	0.4*	mg/L		4		09/14/2023	11:22	ldm	EPA 300.0	09/15/2023	21:40	ldm
Nitrogen, Total as Nitrogen	35.0	0.5	mg/L		1	l	09/26/2023	09:55	sta	Calc.	09/29/2023	16:33	lcr
Nitrate + Nitrite as N	35.0	0.1*	mg/L		4		09/14/2023	11:22	ldm	EPA 300.0	09/15/2023	21:40	ldm
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:33	lcr
Conductivity	1200	1	umhos/cm		1		09/18/2023	16:19	amm	SM 4500-H+B	09/19/2023	04:11	amm
Sulfate Sulfur	14.6	0.17	mg/L		1		09/14/2023	11:22	ldm	EPA 300.0	09/15/2023	02:48	ldm
Solids, Total Dissolved (TDS)	800	20	mg/L		1		09/15/2023	13:20	ctl	SM 2540 C	09/18/2023	11:20	ctl
Calcium	107	1	mg/L		1		09/19/2023	09:20	ejc	EPA 200.7	09/20/2023	17:39	ac
Magnesium	57	1	mg/L		1		09/19/2023	09:20	ejc	EPA 200.7	09/20/2023	17:39	ac
Sodium	47	1	mg/L		1		09/19/2023	09:20	ejc	EPA 200.7	09/20/2023	17:39	ac

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
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Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
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Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
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Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 8
Project : Ed Nunes Dairy

Lab No. : VI 2346219-002
Customer No. : 4019696
Reference : 3155
Sampled On : September 13, 2023 at 06:30
Sampled By : Darin
Received On : September 13, 2023 at 14:42
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	80	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:51	amm
Bicarbonate	90	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:51	amm
Carbonate	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:51	amm
Hydroxide	ND	10	mg/L		1	U	09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:51	amm
Chloride	12	1	mg/L		1		10/11/2023	14:58	ldm	EPA 300.0	10/11/2023	18:52	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:35	lcr
Nitrate Nitrogen	2.9	0.1	mg/L		1		10/11/2023	14:58	ldm	EPA 300.0	10/11/2023	18:52	ldm
Nitrogen, Total as Nitrogen	2.9	0.5	mg/L		1	l	09/26/2023	09:55	sta	Calc.	09/29/2023	16:35	lcr
Nitrate + Nitrite as N	2.9	0.4	mg/L		1		10/11/2023	11:50	lfs	SM 4500-NO3 F	10/11/2023	15:46	lfs
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:35	lcr
Conductivity	237	1	umhos/cm		1		09/20/2023	17:39	amm	SM 4500-H+B	09/21/2023	01:51	amm
Sulfate Sulfur	3.8	0.17	mg/L		1		10/11/2023	14:58	ldm	EPA 300.0	10/11/2023	18:52	ldm
Solids, Total Dissolved (TDS)	180	20	mg/L		1		09/15/2023	11:00	ctl	SM 2540 C	09/18/2023	11:20	ctl
Calcium	18	1	mg/L		1	h	09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:15	ac
Magnesium	10	1	mg/L		1		09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:15	ac
Sodium	17	1	mg/L		1	h	09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:15	ac

DQF Flags Definition:
U Constituent results were non-detect.
l The MS/MSD did not meet QC criteria.
h The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
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CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
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San Luis Obispo, CA 93401
TEL: (805)783-2940
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9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
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October 16, 2023

Sentry Ag Services
Attn: Monique Baldivez
P.O. Box 7750
Visalia, CA 93290

Description : 9
Project : Ed Nunes Dairy

Lab No. : VI 2346219-003
Customer No. : 4019696
Reference : 3155
Sampled On : September 13, 2023 at 06:35
Sampled By : Darin
Received On : September 13, 2023 at 14:42
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Alkalinity (as CaCO3)	240	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	19:18	amm
Bicarbonate	300	10	mg/L		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	19:18	amm
Carbonate	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	19:18	amm
Hydroxide	ND	10	mg/L		1	UI	09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	19:18	amm
Chloride	19	1	mg/L		1		09/14/2023	13:48	ldm	EPA 300.0	09/15/2023	11:11	ldm
Nitrogen, Total Kjeldahl	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:37	lcr
Nitrate Nitrogen	13.2	0.1	mg/L		1		09/14/2023	13:48	ldm	EPA 300.0	09/15/2023	11:11	ldm
Nitrogen, Total as Nitrogen	13.0	0.5	mg/L		1	l	09/26/2023	09:55	sta	Calc.	09/29/2023	16:37	lcr
Nitrate + Nitrite as N	13.0	0.1	mg/L		1		09/14/2023	13:48	ldm	EPA 300.0	09/15/2023	11:11	ldm
Kjeldahl Nitrogen	ND	0.5	mg/L		1	Ul	09/26/2023	09:55	sta	EPA 351.2	09/29/2023	16:37	lcr
Conductivity	655	1	umhos/cm		1		09/20/2023	17:39	amm	SM 4500-H+B	09/20/2023	19:18	amm
Sulfate Sulfur	9.50	0.17	mg/L		1		09/14/2023	13:48	ldm	EPA 300.0	09/15/2023	11:11	ldm
Solids, Total Dissolved (TDS)	430	20	mg/L		1		09/15/2023	13:20	ctl	SM 2540 C	09/18/2023	11:20	ctl
Calcium	61	1	mg/L		1	h	09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:28	ac
Magnesium	29	1	mg/L		1		09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:28	ac
Sodium	28	1	mg/L		1	h	09/15/2023	07:45	ejc	EPA 200.7	09/18/2023	20:28	ac

DQF Flags Definition:
U Constituent results were non-detect.
I The RPD for the laboratory duplicate exceeded laboratory criteria.
l The MS/MSD did not meet QC criteria.
h The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level * RL adjusted for dilution, Dil.=Dilution

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
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Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 16, 2023
Sentry Ag Service

Lab No. : VI 2346219
Customer No. : 4019696

Quality Control - Metals

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Metals								
Calcium	200.7	09/15/2023:210357EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	98.2%	85-115	
			MS	mg/L	12.00	168%	75-125	435
		(CC 2383134-003)	MSD	mg/L	12.00	149%	75-125	435
			MSRPD	mg/L		3.7%	≤20.0	
			MS	mg/L	12.00	230%	<¼	406
		(CC 2383134-001)	MSD	mg/L	12.00	177%	<1/4	
			MSRPD	mg/L		5.6%	≤20.0	
	200.7	09/19/2023:210515EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	104%	85-115	
			MS	mg/L	12.00	105%	75-125	
		(STK2352436-004)	MSD	mg/L	12.00	97.4%	75-125	
			MSRPD	mg/L		3.8%	≤20.0	
			MS	mg/L	12.00	83.6%	75-125	
		(CC 2383155-001)	MSD	mg/L	12.00	54.6%	<1/4	
			MSRPD	mg/L		4.2%	≤20.0	
Magnesium	200.7	09/15/2023:210357EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	98.5%	85-115	
			MS	mg/L	12.00	192%	<¼	406
		(CC 2383134-003)	MSD	mg/L	12.00	173%	<1/4	
			MSRPD	mg/L		2.5%	≤20	
			MS	mg/L	12.00	272%	<¼	406
		(CC 2383134-001)	MSD	mg/L	12.00	225%	<1/4	
			MSRPD	mg/L		4.4%	≤20	
	200.7	09/19/2023:210515EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	105%	85-115	
			MS	mg/L	12.00	88.7%	75-125	
		(STK2352436-004)	MSD	mg/L	12.00	78.6%	75-125	
			MSRPD	mg/L		1.6%	≤20	
			MS	mg/L	12.00	87.5%	75-125	
		(CC 2383155-001)	MSD	mg/L	12.00	59.4%	<1/4	
			MSRPD	mg/L		4.0%	≤20	
Sodium	200.7	09/15/2023:210357EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	97.5%	85-115	
			MS	mg/L	12.00	136%	75-125	435
		(CC 2383134-003)	MSD	mg/L	12.00	134%	75-125	435
			MSRPD	mg/L		0.8%	≤20.0	
			MS	mg/L	12.00	250%	<¼	406
		(CC 2383134-001)	MSD	mg/L	12.00	201%	<1/4	
			MSRPD	mg/L		4.3%	≤20.0	
	200.7	09/19/2023:210515EJC	Blank	mg/L		ND	<1	
			LCS	mg/L	12.00	114%	85-115	
			MS	mg/L	12.00	50.7%	<¼	406
		(STK2352436-004)	MSD	mg/L	12.00	-41.4%	<1/4	
			MSRPD	mg/L		4.0%	≤20.0	
			MS	mg/L	12.00	87.1%	75-125	

Corporate Offices & Laboratory
853 Corporation Street
Santa Paula, CA 93060
TEL: (805)392-2000
Env FAX: (805)525-4172 / Ag FAX: (805)392-2063
CA ELAP Certification No. 1573

Office & Laboratory
2500 Stagecoach Road
Stockton, CA 95215
TEL: (209)942-0182
FAX: (209)942-0423
CA ELAP Certification No. 1563

Office & Laboratory
563 E. Lindo Avenue
Chico, CA 95926
TEL: (530)343-5818
FAX: (530)343-3807
CA ELAP Certification No. 2670

Office & Laboratory
3442 Empresa Drive, Suite D
San Luis Obispo, CA 93401
TEL: (805)783-2940
FAX: (805)783-2912
CA ELAP Certification No. 2775

Office & Laboratory
9415 W. Goshen Avenue
Visalia, CA 93291
TEL: (559)734-9473
FAX: (559)734-8435
CA ELAP Certification No. 2810

October 16, 2023

Sentry Ag Service

Lab No. : VI 2346219
Customer No. : 4019696

Quality Control - Metals

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
		(CC 2383155-001)	MSD	mg/L	12.00	68.5%	<1/4	
			MSRPD	mg/L		3.6%	≤20.0	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

DQO : Data Quality Objective - This is the criteria against which the quality control data is compared.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Explanation

406 : Matrix Spike (MS) not within the Acceptance Range (AR) because of high analyte concentration in the sample. Data was accepted based on the LCS or CCV recovery.

435 : Sample matrix may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

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October 16, 2023
Sentry Ag Service

Lab No. : VI 2346219
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
Wet Chem								
Alkalinity (as CaCO3)	2320B	09/18/2023:210413AMM	ND	mg/L		0.4%	10	435
	2320B	(VI 2346309-001)	Dup	mg/L		6.26%	10	
		(SP 2315465-001)	Dup	mg/L		0.1%	10	
Bicarbonate	2320B	(CH 2377803-001)	Dup	mg/L		0.4%	10	
	2320B	(VI 2346309-001)	Dup	mg/L		8.88%	10	
		(SP 2315465-001)	Dup	mg/L		0.1%	10	
Carbonate	2320B	(CH 2377803-001)	Dup	mg/L			10	
	2320B	(VI 2346309-001)	Dup	mg/L		15.4%	10	440
E. C.	2320B	(CH 2377803-001)	Dup	umhos/cm		0.6%	5	
	2320B	(VI 2346309-001)	Dup	umhos/cm		0%	5	
		(SP 2315465-001)	Dup	umhos/cm		0.2%	5	
Solids, Total Dissolved	2540CE	09/15/2023:210364CTL	Blank	mg/L		ND	<20	
			LCS	mg/L	991.5	100%	90-110	
		(STK2352326-003)	Dup	mg/L		2.88%	5	
		(STK2352326-003)	Dup	mg/L		1.22%	5	
			Blank	mg/L		ND	<20	
			LCS	mg/L	991.5	101%	90-110	
		(VI 2346219-001)	Dup	mg/L		0.6%	5	
		(VI 2346219-001)	Dup	mg/L		2.30%	5	
Chloride	300.0	09/14/2023:210436LDM	Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	102%	90-110	
			MS	mg/L	50.00	90.8%	67-117	
		(SP 2315544-001)	MSD	mg/L	50.00	90.6%	67-117	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	50.00	102%	67-117	
		(SP 2315528-001)	MSD	mg/L	50.00	102%	67-117	
			MSRPD	mg/L		0.0%	≤7	
			Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	103%	90-110	
			MS	mg/L	50.00	102%	67-117	
		(VI 2346194-001)	MSD	mg/L	50.00	102%	67-117	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	50.00	102%	67-117	
		(VI 2346213-002)	MSD	mg/L	50.00	102%	67-117	
			MSRPD	mg/L		0.4%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<1	
			LCS	mg/L	25.00	99.3%	90-110	
			MS	mg/L	50.00	92.5%	67-117	
		(CH 2378529-001)	MSD	mg/L	50.00	92.5%	67-117	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	50.00	91.0%	67-117	
		(CH 2378530-001)	MSD	mg/L	50.00	91.5%	67-117	
			MSRPD	mg/L		0.4%	≤7	
Nitrate + Nitrite as N	300.0	09/14/2023:210436LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	103%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(SP 2315544-001)	MSD	mg/L	40.00	103%	86-112	
			MSRPD	mg/L		0.2%	≤7	
			MS	mg/L	40.00	104%	86-112	
		(SP 2315528-001)	MSD	mg/L	40.00	104%	86-112	

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October 16, 2023
Sentry Ag Service

Lab No. : VI 2346219
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
			MSRPD	mg/L		0.1%	≤7	
			Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	104%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(VI 2346194-001)	MSD	mg/L	40.00	104%	86-112	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	40.00	103%	86-112	
		(VI 2346213-002)	MSD	mg/L	40.00	102%	86-112	
			MSRPD	mg/L		0.3%	≤7	
Nitrate Nitrogen	300.0	09/14/2023:210436LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	103%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(SP 2315544-001)	MSD	mg/L	40.00	103%	86-112	
			MSRPD	mg/L		0.2%	≤7	
			MS	mg/L	40.00	104%	86-112	
		(SP 2315528-001)	MSD	mg/L	40.00	104%	86-112	
			MSRPD	mg/L		0.1%	≤7	
			Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	104%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(VI 2346194-001)	MSD	mg/L	40.00	104%	86-112	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	40.00	103%	86-112	
		(VI 2346213-002)	MSD	mg/L	40.00	102%	86-112	
			MSRPD	mg/L		0.3%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<0.4	
			LCS	mg/L	20.00	99.5%	90-110	
			MS	mg/L	40.00	104%	86-112	
		(CH 2378529-001)	MSD	mg/L	40.00	103%	86-112	
			MSRPD	mg/L		0.2%	≤7	
			MS	mg/L	40.00	102%	86-112	
		(CH 2378530-001)	MSD	mg/L	40.00	102%	86-112	
			MSRPD	mg/L		0.2%	≤7	
Sulfate Sulfur	300.0	09/14/2023:210436LDM	Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	104%	90-110	
			MS	mg/L	100.0	100%	18-165	
		(SP 2315544-001)	MSD	mg/L	100.0	99.9%	18-165	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	100.0	72.3%	18-165	
		(SP 2315528-001)	MSD	mg/L	100.0	81.1%	18-165	
			MSRPD	mg/L		3.9%	≤7	
			Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	105%	90-110	
			MS	mg/L	100.0	103%	18-165	
		(VI 2346194-001)	MSD	mg/L	100.0	103%	18-165	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	100.0	105%	18-165	
		(VI 2346213-002)	MSD	mg/L	100.0	105%	18-165	
			MSRPD	mg/L		0.4%	≤7	
	300.0	10/11/2023:211481LDM	Blank	mg/L		ND	<0.5	
			LCS	mg/L	50.00	99.6%	90-110	

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October 16, 2023

Sentry Ag Service

Lab No. : VI 2346219
Customer No. : 4019696

Quality Control - Wet Chem

Constituent	Method	Date/ID	Type	Units	Conc.	QC Data	DQO	Note
			MS	mg/L	100.0	102%	18-165	
		(CH 2378529-001)	MSD	mg/L	100.0	102%	18-165	
			MSRPD	mg/L		0.1%	≤7	
			MS	mg/L	100.0	99.2%	18-165	
		(CH 2378530-001)	MSD	mg/L	100.0	99.6%	18-165	
			MSRPD	mg/L		0.3%	≤7	
Nitrogen, Total Kjeldahl	351.2	09/26/2023:210792STA	Blank	mg/L		ND	<0.5	
			LCS	mg/L	12.00	91.0%	73-124	
			MS	mg/L	12.00	85.6%	90-110	435
		(VI 2346184-005)	MSD	mg/L	12.00	87.1%	90-110	435
			MSRPD	mg/L		1.7%	≤20	
			MS	mg/L	12.00	86.3%	90-110	435
		(VI 2346184-006)	MSD	mg/L	12.00	89.3%	90-110	435
			MSRPD	mg/L		3.3%	≤20	
Nitrate + Nitrite as N	4500NO3F	10/11/2023:211472LFS	Blank	mg/L		ND	<0.4	
			LCS	mg/L	11.22	100%	80-120	
			MS	mg/L	5.609	98.0%	66-125	
		(CH 2378768-001)	MSD	mg/L	5.609	99.3%	66-125	
			MSRPD	mg/L		1.2%	≤30.4	

Definition

Blank : Method Blank - Prepared to verify that the preparation process is not contributing contamination to the samples.

Dup : Duplicate Sample - A random sample with each batch is prepared and analyzed in duplicate. The relative percent difference is an indication of precision for the preparation and analysis.

LCS : Laboratory Control Standard/Sample - Prepared to verify that the preparation process is not affecting analyte recovery.

MS : Matrix Spikes - A random sample is spiked with a known amount of analyte. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSD : Matrix Spike Duplicate of MS/MSD pair - A random sample duplicate is spiked with a known amount of analyted. The recoveries are an indication of how that sample matrix affects analyte recovery.

MSRPD : MS/MSD Relative Percent Difference (RPD) - The MS relative percent difference is an indication of precision for the preparation and analysis.

ND : Non-detect - Result was below the DQO listed for the analyte.

Explanation

435 : Sample matrix may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

440 : Sample nonhomogeneity may be affecting this analyte. Data was accepted based on the LCS or CCV recovery.

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Laboratory Analysis Work Order

3155

2346219

SITE NAME: Ed Nunes Dairy

LABORATORY: VT / FGL 4-19696

Billing: Sentry Ag Services, LLC
P.O. Box 7750, Visalia, CA 93290

Authorized Copy Release to:
labs@sentryagservices.com

ANALYSIS TO BE COMPLETED

Irrigation/Ground Water (ELAP Standards)

- W1 EC, NO_3N (Dom)
- W2 EC, NO_3N, TDS, TN (Irr)
- W3 NH_4-N (Ammonium)
- W4 EC, NO_3N, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl, TDS (Dom, GM)
- W5 EC, NO_3N, TDS, TN, Ca, Mg, Na, HCO_3, CO_3, SO_4S, Cl (Irr, GM)
- W6 NO_3N, NO_2 (Dom ILRP, Annually)
- W7 Ca, Mg, Na, K, HCO_3, CO_3, SO_4, Cl + Lab Filtering (GWM)
- W8 Other: _____

Plant Tissue

- P1 TN, NO_3N, PO_4P, K (Mid Season - Wheat)
- P2 TN, P, K (Mid-season - Corn)
- P3 TN, TP, TK, Ash, %M (At Harvest)
- P4 TN, %M
- P5 % Moisture
- P6 NIR
- P7 Other: _____

Process Waste Water (lagoon)

- L1 EC, NH_4N, TKN, TP, TK, TDS (Quarterly)
- L2 EC, NO_3N, NH_4N, TKN, TP, TK, TDS, pH (Annually)
- L3 Ca, Mg, Na, HCO_3,CO_3, SO_4S, Cl (Biennially)
- L4 Other: _____

Manure

- M1 TN, TP, TK, %M (2/year)
- M2 TN, TP, K, %M, Ca, Mg, Na, S, Cl, ash (Biennially)
- M3 Other: _____

Soil

- S1 SP%, pH, EC, Ca, Mg, Na, K, ESP, LP, B, NO_3N, PO_4P, K-AA, Zn, Mn, Fe, Cu, SO_4S
- S2 S1 + CEC, CaCO3, OM, C:N, TN
- S3 NO_3N, NH_4N
- S4 Other: _____

	Sample ID	Description	Analysis	Date/Time	Sampled by	SAS USE ONLY: FIELD TESTS NH_3N *	pH	Temp
1	5	irrigation well	W5	9/13/23 6:38	Pedro	8		
2	8	↓	↓	9/13/23 6:30	↓	8		
3	9	↓	↓	9/13/23 6:35	↓	8		
4								
5								
6								
7								
8								
9								
10								
11								
12								

* Field Test of ammonium nitrogen may only be made by a trained technician. Positive test to be analyzed for ammonium nitrogen by the laboratory.

All samples are to follow the procedures noted in the Sampling & Analysis Plan of the NMP and the RWQCB specifications. Any samples taken outside of these procedures shall provide the procedures on the notes below. Additionally, if any preservatives are used in the collections or processing of sampes, please note below.

NOTES:

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	[illegible]	SAS		9/13/23 14:42
2nd	SRO	FGL	9.13.23 1442	
3rd	SRO	FGL		9.13.23 1730
4th	~~GLS~~ SRO 9.13.23	GLS	9.13.23 1730	

LABORATORY USE ONLY
Logged In By: _____ **Total Samples:** _____ **Laboratory No.:** [illegible] 9/14/23 1115.

R01
13.3°C
ID# TH407

Inter-Laboratory Condition Upon Receipt (Attach to COC)

Sample Receipt at: STK CC CH (VI)

1. Number of ice chests/packages received: 1 Shipping tracking # OTC
2. Were samples received in a chilled condition? Temps: RGI / 13.3°C / ____ / ____ / ____
   Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.

ID# TH407

3. Do the number of bottles received agree with the COC?	(Yes)	No	N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.)	(Yes)	No	
5. VOAs checked for Headspace?	Yes	No	(N/A)
6. Were sample custody seals intact?	Yes	No	(N/A)
7. If required, was sample split for pH analysis?	Yes	No	(N/A)
8. Were all analyses within holding times at time of receipt?	(Yes)	No	
9. Verify sample date, time and sampler name	(Yes)	No	

Sign and date the COC, place in a ziplock and put in the same ice chest as the samples.
Sample Receipt Review completed by (initials): SRO

Sample Receipt at SP:

1. Were samples received in a chilled condition? Temps: 3 / 3 / ____ / ____ / ____
   Acceptable is above freezing to 6 C. If many packages are received at one time check for tests/H.T.'s/rushes/
2. Shipping tracking numbers: 5401174957 + 5601113267

3. Do the number of bottles received agree with the COC?	(Yes)	No	N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.)	(Yes)	No	
5. Were sample custody seals intact?	Yes	No	(N/A)

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1. Were all requested analyses understood and acceptable?	(Yes)	No		
2. Did bottle labels correspond with the client's ID's?	(Yes)	No		
3. Were all bottles requiring sample preservation properly preserved? [Exception: Oil & Grease, VOA and CrVI verified in lab]	(Yes)	No	N/A	FGL
4. VOAs checked for Headspace?	Yes	No	(N/A)	
5. Have rush or project due dates been checked and accepted?	(Yes)	No	(N/A)	
6. Were all analyses within holding times at time of receipt?	(Yes)	No		

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): JL

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:

2. Person Contacted: _____
   Initiated By: _____
   Problem:
   Resolution:

(Please use the back of this sheet for additional c
contacts)

(4019696)
Sentry Ag Service
VI 2346219
iv 09/14/2023 08:09:26
VI 2346219